PhonePe Digital Payments Case Study

NAME : THANNIRU DHARMA NITHIN

REGISTERED E-MAIL :thannirudharmanithin2003@gmail.com

Course

Data Analytics – Coding NinjaS

Dataset : phonepe-pulse_raw-data.xlsx

Objective

The objective of this case study is to analyze PhonePe transaction data along with demographic information across various states and districts in India.

The analysis aims to:

• Understand transaction trends across states and years
• Identify device usage patterns among users
• Explore relationships between population demographics and transaction activity
• Perform data quality validation between district and state level datasets
• Generate insights using statistical analysis and visualizations

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Dataset Overview

The dataset contains multiple sheets covering transaction statistics, user behavior, device usage and demographic information across India from **2018 to 2021**.

The datasets used include:

1. State Transactions and Users
2. State Transaction Split
3. State Device Data
4. District Transactions and Users
5. District Demographics

Importing Required Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display settings
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', None)
```

The required Python libraries were imported to perform data analysis and visualization. Pandas is used for handling the dataset, while Matplotlib and Seaborn are used to create visualizations for better understanding of transaction trends.

TASK 1 — DATA LOADING AND UNDERSTANDING

Task 1.1 Load datasets and display structure

The dataset was stored in an Excel file containing multiple sheets. Each sheet represents a different aspect of PhonePe data such as state transactions, transaction categories, device usage, and district level information. These sheets were loaded separately to analyse each dataset effectively.

In [3]:
```
file_path = "phonepe-pulse_raw-data.xlsx"

state_txn_users = pd.read_excel(file_path, sheet_name="State_Txn and Users")
state_txn_split = pd.read_excel(file_path, sheet_name="State_TxnSplit")
state_device_data = pd.read_excel(file_path, sheet_name="State_DeviceData")
district_txn_users = pd.read_excel(file_path, sheet_name="District_Txn and Users
district_demographics = pd.read_excel(file_path, sheet_name="District Demographi
```

1.1.1 Display first 5 rows of State_Txn and Users

In [4]:
```
state_txn_users.head(5)
```

Out[4]:

	State	Year	Quarter	Transactions	Amount (INR)	ATV (INR)	Registered Users	Ap Open
0	Andaman & Nicobar Islands	2018	1	6658	1.463176e+07	2197.621091	6740	
1	Andaman & Nicobar Islands	2018	2	11340	2.833854e+07	2498.989022	9405	
2	Andaman & Nicobar Islands	2018	3	16265	5.555747e+07	3415.768284	12149	
3	Andaman & Nicobar Islands	2018	4	23758	9.054834e+07	3811.277720	15222	
4	Andaman & Nicobar Islands	2019	1	30486	1.022997e+08	3355.630147	18596	

The first few rows of the dataset were displayed to understand the structure of the data and verify that the dataset was loaded correctly.

1.1.2 Display bottom 10 rows of State_TxnSplit

In [5]:
```
state_txn_split.tail(10)
```

Out[5]:

	State	Year	Quarter	Transaction Type	Transactions	Amount (INR)	ATV (INR)
2504	West Bengal	2021	1	Peer-to-peer payments	53869075	2.022402e+11	3754.292226
2505	West Bengal	2021	1	Merchant payments	37143701	2.891834e+10	778.553104
2506	West Bengal	2021	1	Recharge & bill payments	26673733	1.133967e+10	425.124820
2507	West Bengal	2021	1	Financial Services	166727	1.754458e+08	1052.293941
2508	West Bengal	2021	1	Others	400816	2.635025e+08	657.415236
2509	West Bengal	2021	2	Peer-to-peer payments	64661051	2.308123e+11	3569.572026
2510	West Bengal	2021	2	Merchant payments	41696787	3.478787e+10	834.305703
2511	West Bengal	2021	2	Recharge & bill payments	34799709	1.333145e+10	383.090958
2512	West Bengal	2021	2	Financial Services	190537	1.864665e+08	978.636630
2513	West Bengal	2021	2	Others	549353	3.167447e+08	576.577748

Displaying the last 10 rows helps verify the completeness of the dataset and ensures the data has been loaded correctly.

1.1.3 Display middle 10 rows of State_DeviceData

In [7]:
```
mid_index = len(state_device_data) // 2
state_device_data.iloc[mid_index:mid_index+10]
```

Out[7]:

	State	Year	Quarter	Brand	Registered Users	Percentage
2772	Lakshadweep	2018	1	Samsung	102	0.203593
2773	Lakshadweep	2018	1	Xiaomi	100	0.199601
2774	Lakshadweep	2018	1	Vivo	67	0.133733
2775	Lakshadweep	2018	1	Oppo	56	0.111776
2776	Lakshadweep	2018	1	Huawei	25	0.049900
2777	Lakshadweep	2018	1	OnePlus	22	0.043912
2778	Lakshadweep	2018	1	Apple	20	0.039920
2779	Lakshadweep	2018	1	Realme	20	0.039920
2780	Lakshadweep	2018	1	Gionee	12	0.023952
2781	Lakshadweep	2018	1	COOLPAD	10	0.019960

To better understand the dataset, we inspect rows from the middle of the dataset rather than only the beginning or end.

1.1.4 Display first 10 and last 10 rows of District_Txn and Users

In [8]:

```
district_txn_users.head(10)

district_txn_users.tail(10)
```

Out[8]:

	State	Year	Quarter	District	Code	Transactions	Amount (INR)	ATV (
10238	West Bengal	2021	2	Murshidabad	WB14	8602251	1.999694e+10	2324.61
10239	West Bengal	2021	2	Nadia	WB15	5524174	1.122758e+10	2032.44
10240	West Bengal	2021	2	North Twenty Four Parganas	WB16	17258291	3.041482e+10	1762.33
10241	West Bengal	2021	2	Paschim Bardhaman	WB17	4893774	8.731263e+09	1784.15
10242	West Bengal	2021	2	Paschim Medinipur	WB18	5051834	1.030351e+10	2039.55
10243	West Bengal	2021	2	Purba Bardhaman	WB19	3920729	7.572502e+09	1931.40
10244	West Bengal	2021	2	Purba Medinipur	WB20	6418522	1.515507e+10	2361.14
10245	West Bengal	2021	2	Purulia	WB21	1895981	2.790996e+09	1472.05
10246	West Bengal	2021	2	South Twenty Four Parganas	WB22	6661813	1.339853e+10	2011.24
10247	West Bengal	2021	2	Uttar Dinajpur	WB23	2253385	5.564221e+09	2469.27

The first and last records of the district transaction dataset were displayed to verify that the data loaded correctly and to understand the structure of district level transaction information.

1.1.5 Display every 10th row of District Demographics

In [9]:
```
district_demographics.iloc[::10]
```

Out[9]:

	State	District	Headquarters	Population	Area (sq km)	Density	Code
0	Andhra Pradesh	Anantapur	Anantapur	4083315	19130.00	213	AP01
10	Andhra Pradesh	Visakhapatnam	Visakhapatnam	4288113	11161.00	384	AP10
20	Arunachal Pradesh	Lepa Rada	Basar	0	0.00	0	AR08
30	Arunachal Pradesh	Siang	Pangin	31920	2919.00	11	AR18
40	Assam	Barpeta	Barpeta	1693622	3245.00	520	AS03
50	Assam	Dima Hasao	Haflong	213529	4888.00	44	AS13
60	Assam	Kokrajhar	Kokrajhar	886999	3129.00	283	AS23
70	Assam	Udalguri	Udalguri	832769	1676.00	497	AS33
80	Bihar	Darbhanga	Darbhanga	3921971	2278.00	1721	BR09
90	Bihar	Lakhisarai	Lakhisarai	1000717	1229.00	815	BR19
100	Bihar	Saharsa	Saharsa	1897102	1702.00	1125	BR29
110	Chhattisgarh	Balod	Balod	826165	3527.00	234	CG01
120	Chhattisgarh	Gariaband	Gariaband	597653	5823.00	103	CG11
130	Chhattisgarh	Mungeli	Mungeli	701707	2750.00	255	CG21
140	Gujarat	Ahmedabad	Ahmedabad	7208200	8707.00	890	GJ01
150	Gujarat	Dang	Ahwa	226769	1764.00	129	GJ11
160	Gujarat	Morbi	Morbi	960329	4871.00	197	GJ21
170	Gujarat	Tapi	Vyara	806489	3435.00	249	GJ31
180	Haryana	Jhajjar	Jhajjar	956907	1868.00	522	HR08
190	Haryana	Rewari	Rewari	896129	1559.00	562	HR18
200	Himachal Pradesh	Kullu	Kullu	437474	5503.00	79	HP06
210	Jharkhand	Dhanbad	Dhanbad	2682662	2075.00	1284	JH04
220	Jharkhand	Koderma	Koderma	717169	1312.00	427	JH14
230	Jharkhand	West Singhbhum	Chaibasa	1501619	7186.00	209	JH24
240	Karnataka	Chikmagalur	Chikmagalur	1137753	7201.00	158	KA10
250	Karnataka	Kolar	Kolar	1540231	4012.00	384	KA20
260	Karnataka	Vijayanagara	Hospete	1353628	5644.00	240	KA30
270	Kerala	Malappuram	Malappuram	4110956	3554.00	1058	KL09

	State	District	Headquarters	Population	Area (sq km)	Density	Code
280	Madhya Pradesh	Balaghat	Balaghat	1701156	9229.00	184	MP05
290	Madhya Pradesh	Datia	Datia	786375	2694.00	292	MP15
300	Madhya Pradesh	Jhabua	Jhabua	1024091	6782.00	285	MP25
310	Madhya Pradesh	Neemuch	Neemuch	825958	4267.00	194	MP35
320	Madhya Pradesh	Seoni	Seoni	1378876	8758.00	157	MP45
330	Madhya Pradesh	Vidisha	Vidisha	1458212	7362.00	198	MP55
340	Maharashtra	Gadchiroli	Gadchiroli	1071795	14412.00	74	MH10
350	Maharashtra	Nandurbar	Nandurbar	1646177	5055.00	276	MH21
360	Maharashtra	Satara	Satara	3003922	10475.00	287	MH30
370	Manipur	Imphal East	Porompat	452661	710.00	555	MN04
380	Manipur	Tengnoupal	Tengnoupal	0	0.00	0	MN14
390	Meghalaya	South West Khasi Hills	Mawkyrwat	110152	1341.00	82	ML08
400	Mizoram	Lunglei	Lunglei	154094	4572.00	34	MZ07
410	Nagaland	Mon	Mon	259604	1786.00	145	NL06
420	Odisha	Balangir	Balangir	1648574	6575.00	251	OR02
430	Odisha	Jagatsinghpur	Jagatsinghpur	1136604	1759.00	681	OR12
440	Odisha	Nuapada	Nuapada	606490	3408.00	157	OR25
450	Punjab	Firozpur	Firozpur	2026831	5334.00	380	PB07
460	Punjab	Mansa	Mansa	768808	2174.00	350	PB14
470	Rajasthan	Ajmer	Ajmer	2584913	8481.00	305	RJ01
480	Rajasthan	Chittorgarh	Chittorgarh	1544392	10856.00	193	RJ10
490	Rajasthan	Jaisalmer	Jaisalmer	672008	38401.00	17	RJ17
500	Rajasthan	Sirohi	Sirohi	1037185	5136.00	202	RJ30
510	Tamil Nadu	Coimbatore	Coimbatore	3472578	4723.00	748	TN04
520	Tamil Nadu	Madurai	Madurai	3191038	3676.00	823	TN14
530	Tamil Nadu	Sivaganga	Sivaganga	1341250	4086.00	324	TN24
540	Tamil Nadu	Tiruvarur	Thiruvarur	1268094	2377.00	533	TN35
550	Telangana	Jangaon	Jangaon	582457	2187.50	259	TG05

	State	District	Headquarters	Population	Area (sq km)	Density	Code
560	Telangana	Medchal–Malkajgiri	Shamirpet	2542203	5005.98	2251	TG16
570	Telangana	Sangareddy	Sangareddy	1527628	4464.87	347	TG26
580	Tripura	Khowai	Khowai	327391	1005.67	326	TR03
590	Uttar Pradesh	Amethi	Gauriganj	2549935	3063.00	830	UP05
600	Uttar Pradesh	Bareilly	Bareilly	4465344	4120.00	1084	UP15
610	Uttar Pradesh	Etawah	Etawah	1579160	2287.00	683	UP25
620	Uttar Pradesh	Hamirpur	Hamirpur	1104021	4325.00	268	UP35
630	Uttar Pradesh	Kasganj	Kasganj	1438156	1955.00	736	UP45
640	Uttar Pradesh	Mau	Mau	2205170	1713.00	1287	UP55
650	Uttar Pradesh	Sambhal	Sambhal	2217020	2453.00	890	UP65
660	Uttar Pradesh	Varanasi	Varanasi	3682194	1535.00	2399	UP75
670	Uttarakhand	Rudraprayag	Rudraprayag	236857	1896.00	119	UK10
680	West Bengal	Dakshin Dinajpur	Balurghat	1670931	2183.00	753	WB05
690	West Bengal	Nadia	Krishnanagar	5168488	3927.00	1316	WB15
700	Chandigarh	Chandigarh	Chandigarh	1055450	114.00	9258	CH01
710	Jammu & Kashmir	Jammu	Jammu	1526406	3097.00	596	JK07
720	Jammu & Kashmir	Samba	Samba	318611	913.00	318	JK17
730	Delhi	North Delhi	Sadar Bazaar	887978	59.00	14557	DL04
740	Puducherry	Puducherry	Pondicherry	950289	293.00	3232	PY03

Displaying every 10th row provides a quick overview of the dataset distribution across districts.

Task 1.2 Summary Statistics and Data Types

In [10]:
```
state_txn_users.describe()
```

Out[10]:

	Year	Quarter	Transactions	Amount (INR)	ATV (INR)	Registered Users
count	504.000000	504.000000	5.040000e+02	5.030000e+02	504.000000	5.040000e+02
mean	2019.285714	2.357143	4.074000e+07	7.083332e+10	1993.471543	4.777501e+06
std	1.031181	1.109971	8.228714e+07	1.440902e+11	607.464894	6.644496e+06
min	2018.000000	1.000000	7.780000e+02	1.928611e+06	0.000000	5.010000e+02
25%	2018.000000	1.000000	5.925578e+05	1.167157e+09	1598.910667	1.574202e+05
50%	2019.000000	2.000000	6.217487e+06	1.051605e+10	1861.380589	1.747914e+06
75%	2020.000000	3.000000	4.363675e+07	6.947045e+10	2259.087924	7.320945e+06
max	2021.000000	4.000000	5.736165e+08	1.027958e+12	3938.733850	3.966470e+07

In [11]:
```
state_txn_users.dtypes
```

Out[11]:
```
State                object
Year                  int64
Quarter               int64
Transactions          int64
Amount (INR)        float64
ATV (INR)           float64
Registered Users      int64
App Opens             int64
dtype: object
```

In [12]:
```
state_txn_split.describe()
state_txn_split.dtypes

state_device_data.describe()
state_device_data.dtypes

district_txn_users.describe()
district_txn_users.dtypes

district_demographics.describe()
district_demographics.dtypes
```

Out[12]:
```
State              object
District           object
Headquarters       object
Population          int64
Area (sq km)      float64
Density             int64
Code               object
Alternate Name     object
dtype: object
```

Summary statistics and data type checks were performed on all datasets to understand their structure and ensure data consistency before performing further analysis.

Task 1.3 Checking Missing Values

In [14]:
```
def missing_values(df):
    missing = df.isnull().sum()
    percent = (missing/len(df))*100
    missing_table = pd.DataFrame({
        "Missing Values": missing,
        "Percentage": percent
    })
    return missing_table[missing_table["Missing Values"] > 0]
```

In [15]:
```
missing_values(state_txn_users)
missing_values(state_txn_split)
missing_values(state_device_data)
missing_values(district_txn_users)
missing_values(district_demographics)
```

Out[15]:

Missing Values	Percentage

A custom function was created to identify missing values in the datasets. The function calculates both the number and percentage of missing values for each column.

Task 1.4 Dataset Summary

In [16]:
```
total_states = state_txn_users["State"].nunique()
print("Total number of states:", total_states)
```

```
Total number of states: 36
```

In [17]:
```
total_districts = district_demographics["District"].nunique()
print("Total number of districts:", total_districts)
```

```
Total number of districts: 736
```

In [18]:
```
district_count = district_demographics["State"].value_counts()

state_max_districts = district_count.idxmax()

print("State with highest number of districts:", state_max_districts)
```

```
State with highest number of districts: Uttar Pradesh
```

This summary provides a quick overview of the dataset coverage and helps identify states with the highest number of administrative districts.

Task 1 Summary

In this section we successfully loaded all datasets and performed an initial exploration.

Key activities included:

- Loading datasets from Excel
- Inspecting sample records
- Analyzing data types and summary statistics

• Identifying missing values
• Calculating number of states and districts

This step helps ensure the dataset is properly understood before performing deeper analysis.

Task 2: Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand patterns, trends, and relationships within the dataset.

In this section we analyze:

- Transaction trends across states
- Most common transaction types
- Device brand usage
- District population patterns
- Average transaction values
- App usage trends
- Distribution of transaction types

Task 2.1 Analyze Transaction Trends Across States

State level transaction totals were calculated by grouping the data by state and summing the number of transactions and transaction amounts.

In [19]:
```
txn_trend = state_txn_users.groupby("State")[["Transactions","Amount (INR)"]].su

txn_trend
```

Out[19]:

	State	Transactions	Amount (INR)
0	Andaman & Nicobar Islands	1223565	3.641818e+09
1	Andhra Pradesh	1781091169	3.187847e+12
2	Arunachal Pradesh	8043799	2.225183e+10
3	Assam	123577981	2.814087e+11
4	Bihar	713643889	1.639579e+12
5	Chandigarh	25516542	5.371208e+10
6	Chhattisgarh	241908549	4.156353e+11
7	Dadra & Nagar Haveli and Daman & Diu	12462185	1.865878e+10
8	Delhi	1011031124	1.649942e+12
9	Goa	24481604	4.427187e+10
10	Gujarat	612992823	1.105329e+12
11	Haryana	567751410	1.059832e+12
12	Himachal Pradesh	52987445	9.786546e+10
13	Jammu & Kashmir	45053328	1.094954e+11
14	Jharkhand	275270987	4.905701e+11
15	Karnataka	2981044533	4.361396e+12
16	Kerala	223637876	3.729599e+11
17	Ladakh	1880109	6.606978e+09
18	Lakshadweep	71610	1.921587e+08
19	Madhya Pradesh	1100253728	1.915101e+12
20	Maharashtra	2833670154	4.290995e+12
21	Manipur	12318043	3.579325e+10
22	Meghalaya	5648913	1.254162e+10
23	Mizoram	2162776	6.317141e+09
24	Nagaland	5864977	1.638757e+10
25	Odisha	740479146	1.220615e+12
26	Puducherry	17292386	3.209769e+10
27	Punjab	181336904	3.941773e+11
28	Rajasthan	1382918930	2.625584e+12
29	Sikkim	6483948	1.415554e+10
30	Tamil Nadu	800399873	1.481448e+12
31	Telangana	2347430243	4.574246e+12
32	Tripura	12018152	2.097051e+10

	State	Transactions	Amount (INR)
33	Uttar Pradesh	1314714390	2.399757e+12
34	Uttarakhand	124011162	2.150715e+11
35	West Bengal	942286041	1.452707e+12

In [20]:
```
top5_states = txn_trend.sort_values(by="Transactions",ascending=False).head(5)

top5_states
```

Out[20]:

	State	Transactions	Amount (INR)
15	Karnataka	2981044533	4.361396e+12
20	Maharashtra	2833670154	4.290995e+12
31	Telangana	2347430243	4.574246e+12
1	Andhra Pradesh	1781091169	3.187847e+12
28	Rajasthan	1382918930	2.625584e+12

In [21]:
```
bottom5_states = txn_trend.sort_values(by="Transactions").head(5)

bottom5_states
```

Out[21]:

	State	Transactions	Amount (INR)
18	Lakshadweep	71610	1.921587e+08
0	Andaman & Nicobar Islands	1223565	3.641818e+09
17	Ladakh	1880109	6.606978e+09
23	Mizoram	2162776	6.317141e+09
22	Meghalaya	5648913	1.254162e+10

Task 2.2 Most Frequent Transaction Types

In [23]:
```
most_common_txn = state_txn_split.loc[
    state_txn_split.groupby(["State","Quarter"])["Transactions"].idxmax()
]

most_common_txn[["State","Quarter","Transaction Type","Transactions"]]
```

Out[23]:

	State	Quarter	Transaction Type	Transactions
60	Andaman & Nicobar Islands	1	Peer-to-peer payments	105458
65	Andaman & Nicobar Islands	2	Peer-to-peer payments	132772
50	Andaman & Nicobar Islands	3	Recharge & bill payments	59310
55	Andaman & Nicobar Islands	4	Peer-to-peer payments	81439
130	Andhra Pradesh	1	Peer-to-peer payments	149313738
135	Andhra Pradesh	2	Peer-to-peer payments	189614879
120	Andhra Pradesh	3	Peer-to-peer payments	106791894
125	Andhra Pradesh	4	Peer-to-peer payments	152003791
200	Arunachal Pradesh	1	Peer-to-peer payments	566209
205	Arunachal Pradesh	2	Peer-to-peer payments	762946
190	Arunachal Pradesh	3	Recharge & bill payments	514907
195	Arunachal Pradesh	4	Peer-to-peer payments	522019
270	Assam	1	Peer-to-peer payments	9449096
275	Assam	2	Peer-to-peer payments	11668399
260	Assam	3	Peer-to-peer payments	5637893
265	Assam	4	Peer-to-peer payments	7538478
340	Bihar	1	Peer-to-peer payments	58566582
345	Bihar	2	Peer-to-peer payments	78694099
330	Bihar	3	Peer-to-peer payments	37294332
335	Bihar	4	Peer-to-peer payments	49432098
410	Chandigarh	1	Merchant payments	1799072
415	Chandigarh	2	Merchant payments	1926913
400	Chandigarh	3	Peer-to-peer payments	981986
405	Chandigarh	4	Merchant payments	1322026
480	Chhattisgarh	1	Peer-to-peer payments	15621154
485	Chhattisgarh	2	Peer-to-peer payments	18387345
470	Chhattisgarh	3	Peer-to-peer payments	9734445
475	Chhattisgarh	4	Peer-to-peer payments	14257312
550	Dadra & Nagar Haveli and Daman & Diu	1	Peer-to-peer payments	768921
555	Dadra & Nagar Haveli and Daman & Diu	2	Peer-to-peer payments	843550

	State	Quarter	Transaction Type	Transactions
540	Dadra & Nagar Haveli and Daman & Diu	3	Recharge & bill payments	455603
545	Dadra & Nagar Haveli and Daman & Diu	4	Peer-to-peer payments	666995
620	Delhi	1	Merchant payments	72615587
625	Delhi	2	Merchant payments	75620295
610	Delhi	3	Peer-to-peer payments	41498638
615	Delhi	4	Merchant payments	54581166
690	Goa	1	Peer-to-peer payments	2235456
695	Goa	2	Merchant payments	1869384
680	Goa	3	Peer-to-peer payments	759351
685	Goa	4	Peer-to-peer payments	1550251
760	Gujarat	1	Peer-to-peer payments	36427493
765	Gujarat	2	Merchant payments	42593432
750	Gujarat	3	Peer-to-peer payments	23130724
755	Gujarat	4	Peer-to-peer payments	29967258
830	Haryana	1	Merchant payments	39363588
835	Haryana	2	Merchant payments	45335533
820	Haryana	3	Peer-to-peer payments	24123449
825	Haryana	4	Merchant payments	34063692
900	Himachal Pradesh	1	Peer-to-peer payments	2845733
905	Himachal Pradesh	2	Peer-to-peer payments	3340531
890	Himachal Pradesh	3	Recharge & bill payments	2416395
895	Himachal Pradesh	4	Recharge & bill payments	2638030
970	Jammu & Kashmir	1	Peer-to-peer payments	2800201
975	Jammu & Kashmir	2	Peer-to-peer payments	3955437
960	Jammu & Kashmir	3	Peer-to-peer payments	1787154
965	Jammu & Kashmir	4	Peer-to-peer payments	2378539
1040	Jharkhand	1	Peer-to-peer payments	19151083
1045	Jharkhand	2	Peer-to-peer payments	24184568
1030	Jharkhand	3	Peer-to-peer payments	11984094
1035	Jharkhand	4	Peer-to-peer payments	17120664

	State	Quarter	Transaction Type	Transactions
1110	Karnataka	1	Merchant payments	279599017
1115	Karnataka	2	Merchant payments	244460862
1100	Karnataka	3	Merchant payments	137206600
1105	Karnataka	4	Merchant payments	215711211
1180	Kerala	1	Peer-to-peer payments	16209203
1185	Kerala	2	Peer-to-peer payments	17026648
1170	Kerala	3	Peer-to-peer payments	8590587
1175	Kerala	4	Peer-to-peer payments	12941135
1250	Ladakh	1	Peer-to-peer payments	141894
1255	Ladakh	2	Peer-to-peer payments	209818
1240	Ladakh	3	Peer-to-peer payments	108345
1245	Ladakh	4	Peer-to-peer payments	151582
1314	Lakshadweep	1	Peer-to-peer payments	3942
1319	Lakshadweep	2	Recharge & bill payments	5499
1304	Lakshadweep	3	Recharge & bill payments	4696
1309	Lakshadweep	4	Recharge & bill payments	5187
1384	Madhya Pradesh	1	Merchant payments	84812245
1389	Madhya Pradesh	2	Peer-to-peer payments	99367521
1374	Madhya Pradesh	3	Peer-to-peer payments	50871564
1379	Madhya Pradesh	4	Peer-to-peer payments	67972167
1454	Maharashtra	1	Merchant payments	253043309
1459	Maharashtra	2	Merchant payments	247194293
1444	Maharashtra	3	Peer-to-peer payments	108436214
1449	Maharashtra	4	Merchant payments	189845640
1524	Manipur	1	Peer-to-peer payments	1117829
1529	Manipur	2	Peer-to-peer payments	1372821
1514	Manipur	3	Peer-to-peer payments	774786
1519	Manipur	4	Peer-to-peer payments	892853
1594	Meghalaya	1	Peer-to-peer payments	326993
1599	Meghalaya	2	Recharge & bill payments	389959

	State	Quarter	Transaction Type	Transactions
1584	Meghalaya	3	Recharge & bill payments	342484
1589	Meghalaya	4	Recharge & bill payments	350006
1664	Mizoram	1	Peer-to-peer payments	156386
1669	Mizoram	2	Peer-to-peer payments	173064
1654	Mizoram	3	Recharge & bill payments	132664
1659	Mizoram	4	Peer-to-peer payments	141038
1734	Nagaland	1	Peer-to-peer payments	477346
1739	Nagaland	2	Peer-to-peer payments	526966
1724	Nagaland	3	Recharge & bill payments	347494
1729	Nagaland	4	Peer-to-peer payments	417393
1804	Odisha	1	Peer-to-peer payments	54417515
1809	Odisha	2	Peer-to-peer payments	66929109
1794	Odisha	3	Peer-to-peer payments	34182172
1799	Odisha	4	Peer-to-peer payments	47588728
1874	Puducherry	1	Peer-to-peer payments	1406958
1879	Puducherry	2	Peer-to-peer payments	1581221
1864	Puducherry	3	Peer-to-peer payments	855511
1869	Puducherry	4	Peer-to-peer payments	1211359
1944	Punjab	1	Peer-to-peer payments	10951083
1949	Punjab	2	Peer-to-peer payments	13180933
1934	Punjab	3	Peer-to-peer payments	7354451
1939	Punjab	4	Peer-to-peer payments	9609405
2014	Rajasthan	1	Peer-to-peer payments	116733195
2019	Rajasthan	2	Peer-to-peer payments	139277445
2004	Rajasthan	3	Peer-to-peer payments	73247856
2009	Rajasthan	4	Peer-to-peer payments	88730951
2084	Sikkim	1	Peer-to-peer payments	403703
2089	Sikkim	2	Peer-to-peer payments	473141
2074	Sikkim	3	Recharge & bill payments	298885
2059	Sikkim	4	Peer-to-peer payments	508471

	State	Quarter	Transaction Type	Transactions
2154	Tamil Nadu	1	Peer-to-peer payments	57446279
2159	Tamil Nadu	2	Peer-to-peer payments	60812902
2144	Tamil Nadu	3	Peer-to-peer payments	31754077
2149	Tamil Nadu	4	Peer-to-peer payments	47800741
2224	Telangana	1	Merchant payments	185230619
2229	Telangana	2	Peer-to-peer payments	230861833
2214	Telangana	3	Peer-to-peer payments	116668399
2219	Telangana	4	Peer-to-peer payments	160607487
2294	Tripura	1	Peer-to-peer payments	643366
2299	Tripura	2	Peer-to-peer payments	703728
2284	Tripura	3	Recharge & bill payments	627363
2289	Tripura	4	Peer-to-peer payments	650386
2434	Uttar Pradesh	1	Peer-to-peer payments	85539321
2439	Uttar Pradesh	2	Peer-to-peer payments	102927942
2424	Uttar Pradesh	3	Peer-to-peer payments	54449911
2429	Uttar Pradesh	4	Peer-to-peer payments	75433782
2364	Uttarakhand	1	Merchant payments	8393767
2369	Uttarakhand	2	Peer-to-peer payments	9374939
2354	Uttarakhand	3	Peer-to-peer payments	4836512
2359	Uttarakhand	4	Peer-to-peer payments	6748183
2504	West Bengal	1	Peer-to-peer payments	53869075
2509	West Bengal	2	Peer-to-peer payments	64661051
2454	West Bengal	3	Peer-to-peer payments	41882429
2479	West Bengal	4	Peer-to-peer payments	67037576

Interpretation

This analysis reveals which transaction categories dominate digital payments.

Typically, Peer-to-Peer payments and Recharge/Bill Payments contribute the largest share.

Task 2.3 Device Brand With Highest Registered Users

Here we identify the most popular smartphone brand used to access the PhonePe application in each state.

In [24]:
```
top_device_brand = state_device_data.loc[
    state_device_data.groupby("State")["Registered Users"].idxmax()
]

top_device_brand[["State","Brand","Registered Users"]]
```

Out[24]:

	State	Brand	Registered Users
143	Andaman & Nicobar Islands	Vivo	15056
297	Andhra Pradesh	Xiaomi	4937684
451	Arunachal Pradesh	Vivo	63439
605	Assam	Xiaomi	909274
759	Bihar	Xiaomi	4268361
913	Chandigarh	Xiaomi	101761
1067	Chhattisgarh	Vivo	1169224
1221	Dadra & Nagar Haveli and Daman & Diu	Vivo	87001
1375	Delhi	Xiaomi	2731995
1529	Goa	Xiaomi	152107
1683	Gujarat	Vivo	3834323
1837	Haryana	Xiaomi	2505880
1991	Himachal Pradesh	Xiaomi	513196
2145	Jammu & Kashmir	Xiaomi	446949
2299	Jharkhand	Xiaomi	1595475
2453	Karnataka	Xiaomi	6567268
2607	Kerala	Xiaomi	1714598
2761	Ladakh	Xiaomi	21227
2915	Lakshadweep	Vivo	1118
3069	Madhya Pradesh	Xiaomi	4052668
3223	Maharashtra	Xiaomi	9764252
3377	Manipur	Xiaomi	159031
3531	Meghalaya	Xiaomi	48812
3685	Mizoram	Xiaomi	29025
3839	Nagaland	Xiaomi	69897
3993	Odisha	Xiaomi	2776465
4147	Puducherry	Xiaomi	83067
4301	Punjab	Xiaomi	1380642
4455	Rajasthan	Vivo	4603263
4609	Sikkim	Samsung	55452
4763	Tamil Nadu	Xiaomi	4159447
4917	Telangana	Xiaomi	4736356
5071	Tripura	Xiaomi	116546

	State	Brand	Registered Users
5379	Uttar Pradesh	Xiaomi	8037309
5225	Uttarakhand	Xiaomi	681400
5533	West Bengal	Xiaomi	4654760

Interpretation

The result highlights the dominant smartphone brands across states.

Budget Android devices often dominate due to wider accessibility in developing markets.

Task 2.4 District with Highest Population per State

In this section we identify the district with the highest population in each state.

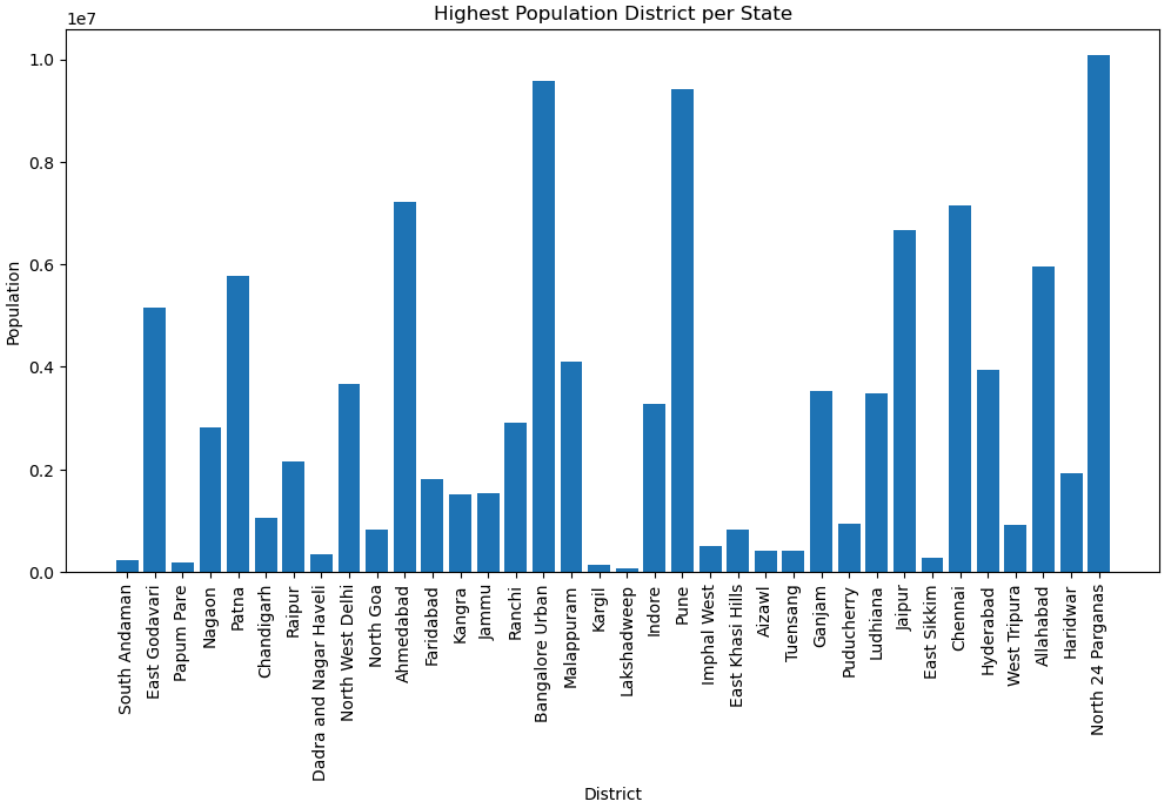
In [25]:
```
top_district_population = district_demographics.loc[
    district_demographics.groupby("State")["Population"].idxmax()
]

top_district_population[["State","District","Population"]]
```

Out[25]:

	State	District	Population
699	Andaman & Nicobar Islands	South Andaman	238142
2	Andhra Pradesh	East Godavari	5151549
28	Arunachal Pradesh	Papum Pare	176385
64	Assam	Nagaon	2826006
97	Bihar	Patna	5772804
700	Chandigarh	Chandigarh	1055450
133	Chhattisgarh	Raipur	2160876
703	Dadra & Nagar Haveli and Daman & Diu	Dadra and Nagar Haveli	343709
732	Delhi	North West Delhi	3656539
138	Goa	North Goa	817761
140	Gujarat	Ahmedabad	7208200
176	Haryana	Faridabad	1798954
198	Himachal Pradesh	Kangra	1507223
710	Jammu & Kashmir	Jammu	1526406
226	Jharkhand	Ranchi	2912022
235	Karnataka	Bangalore Urban	9588910
270	Kerala	Malappuram	4110956
724	Ladakh	Kargil	143388
726	Lakshadweep	Lakshadweep	64473
298	Madhya Pradesh	Indore	3272335
356	Maharashtra	Pune	9426959
371	Manipur	Imphal West	514683
384	Meghalaya	East Khasi Hills	824059
394	Mizoram	Aizawl	404054
414	Nagaland	Tuensang	414801
426	Odisha	Ganjam	3520151
740	Puducherry	Puducherry	950289
458	Punjab	Ludhiana	3487882
489	Rajasthan	Jaipur	6663971
503	Sikkim	East Sikkim	281293
509	Tamil Nadu	Chennai	7139882
548	Telangana	Hyderabad	3943323
585	Tripura	West Tripura	917534

	State	District	Population
588	Uttar Pradesh	Allahabad	5959798
666	Uttarakhand	Haridwar	1927029
691	West Bengal	North 24 Parganas	10082852

In [26]:
```
plt.figure(figsize=(12,6))

plt.bar(
    top_district_population["District"],
    top_district_population["Population"]
)

plt.xticks(rotation=90)

plt.title("Highest Population District per State")

plt.xlabel("District")

plt.ylabel("Population")

plt.show()
```



Interpretation

The chart shows the most populated district in each state.

Task 2.5 Average Transaction Value by State

Average Transaction Value (ATV) indicates the typical size of transactions occurring in each state.

In [27]:
```
avg_atv = state_txn_users.groupby("State")["ATV (INR)"].mean().reset_index()

avg_atv
```

Out[27]:

	State	ATV (INR)
0	Andaman & Nicobar Islands	3159.266422
1	Andhra Pradesh	1748.091476
2	Arunachal Pradesh	2638.323985
3	Assam	2008.495656
4	Bihar	2023.316978
5	Chandigarh	2113.797075
6	Chhattisgarh	1650.818617
7	Dadra & Nagar Haveli and Daman & Diu	1487.395599
8	Delhi	1589.839307
9	Goa	1895.258549
10	Gujarat	1696.101220
11	Haryana	1831.840325
12	Himachal Pradesh	1825.338700
13	Jammu & Kashmir	2508.506029
14	Jharkhand	1632.342997
15	Karnataka	1517.392249
16	Kerala	1671.443173
17	Ladakh	3408.222984
18	Lakshadweep	2776.118701
19	Madhya Pradesh	1601.173668
20	Maharashtra	1516.361820
21	Manipur	2621.225756
22	Meghalaya	2056.276339
23	Mizoram	2927.517802
24	Nagaland	2609.942590
25	Odisha	1428.137367
26	Puducherry	1927.611843
27	Punjab	2085.225325
28	Rajasthan	1757.680992
29	Sikkim	2087.843818
30	Tamil Nadu	1861.374354
31	Telangana	1842.949086
32	Tripura	1554.156042

	State	ATV (INR)
33	Uttar Pradesh	1659.379431
34	Uttarakhand	1672.608685
35	West Bengal	1373.600590

In [32]:
```
highest_atv = avg_atv.sort_values(by="ATV (INR)",ascending=False).head(5)

highest_atv

# Top 5 states with highest ATV
```

Out[32]:

	State	ATV (INR)
17	Ladakh	3408.222984
0	Andaman & Nicobar Islands	3159.266422
23	Mizoram	2927.517802
18	Lakshadweep	2776.118701
2	Arunachal Pradesh	2638.323985

In [34]:
```
lowest_atv = avg_atv.sort_values(by="ATV (INR)").head(5)

lowest_atv
```

Out[34]:

	State	ATV (INR)
35	West Bengal	1373.600590
25	Odisha	1428.137367
7	Dadra & Nagar Haveli and Daman & Diu	1487.395599
20	Maharashtra	1516.361820
15	Karnataka	1517.392249

Interpretation

States with higher ATV values indicate larger financial transactions, which may reflect stronger economic activity or higher-value payments.

Task 2.6 App Usage Trends

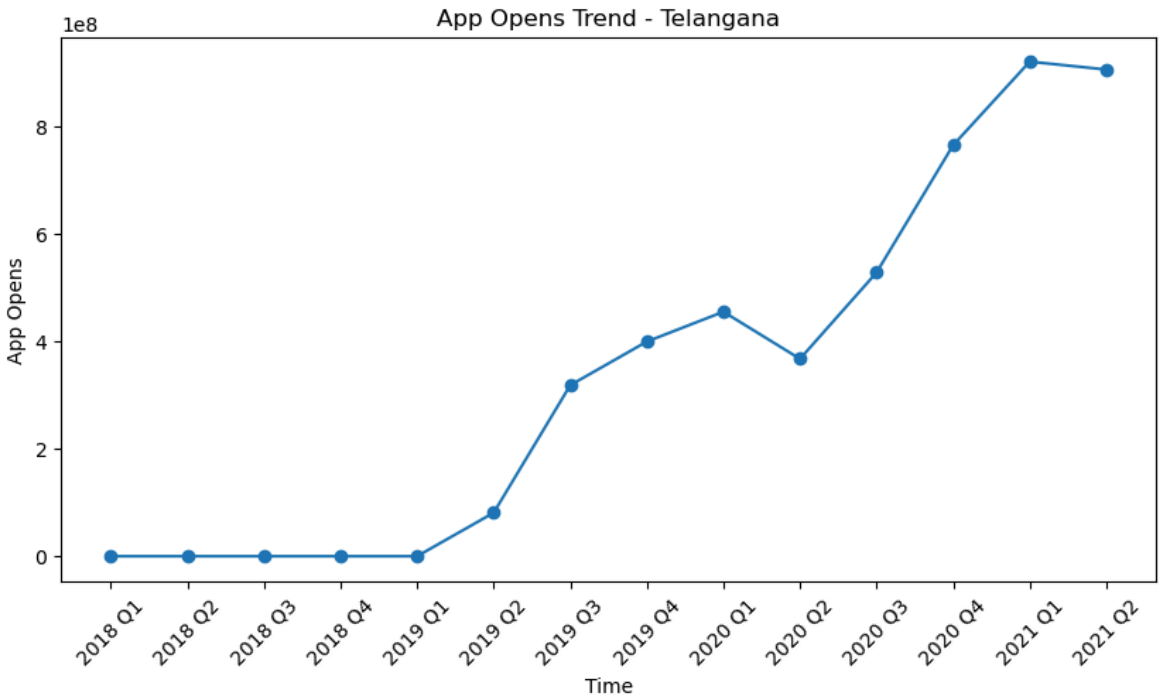
In [35]:
```
app_usage = state_txn_users.groupby(["Year","Quarter"])["App Opens"].sum().reset

app_usage
```

Out[35]:

	Year	Quarter	App Opens
0	2018	1	0
1	2018	2	0
2	2018	3	0
3	2018	4	0
4	2019	1	0
5	2019	2	920409531
6	2019	3	3448355858
7	2019	4	4301061532
8	2020	1	4768934587
9	2020	2	4357622260
10	2020	3	5739306412
11	2020	4	7461591732
12	2021	1	8635472151
13	2021	2	9630577372

In [36]:

```
state_selected = "Telangana"

state_app_usage = state_txn_users[state_txn_users["State"] == state_selected]

usage_trend = state_app_usage.groupby(["Year","Quarter"])["App Opens"].sum().res

usage_trend["Time"] = usage_trend["Year"].astype(str) + " Q" + usage_trend["Quar

plt.figure(figsize=(10,5))

plt.plot(usage_trend["Time"], usage_trend["App Opens"], marker="o")

plt.xticks(rotation=45)

plt.title("App Opens Trend - Telangana")

plt.xlabel("Time")

plt.ylabel("App Opens")

plt.show()
```



Interpretation

The trend line shows how frequently users interact with the PhonePe application over time.

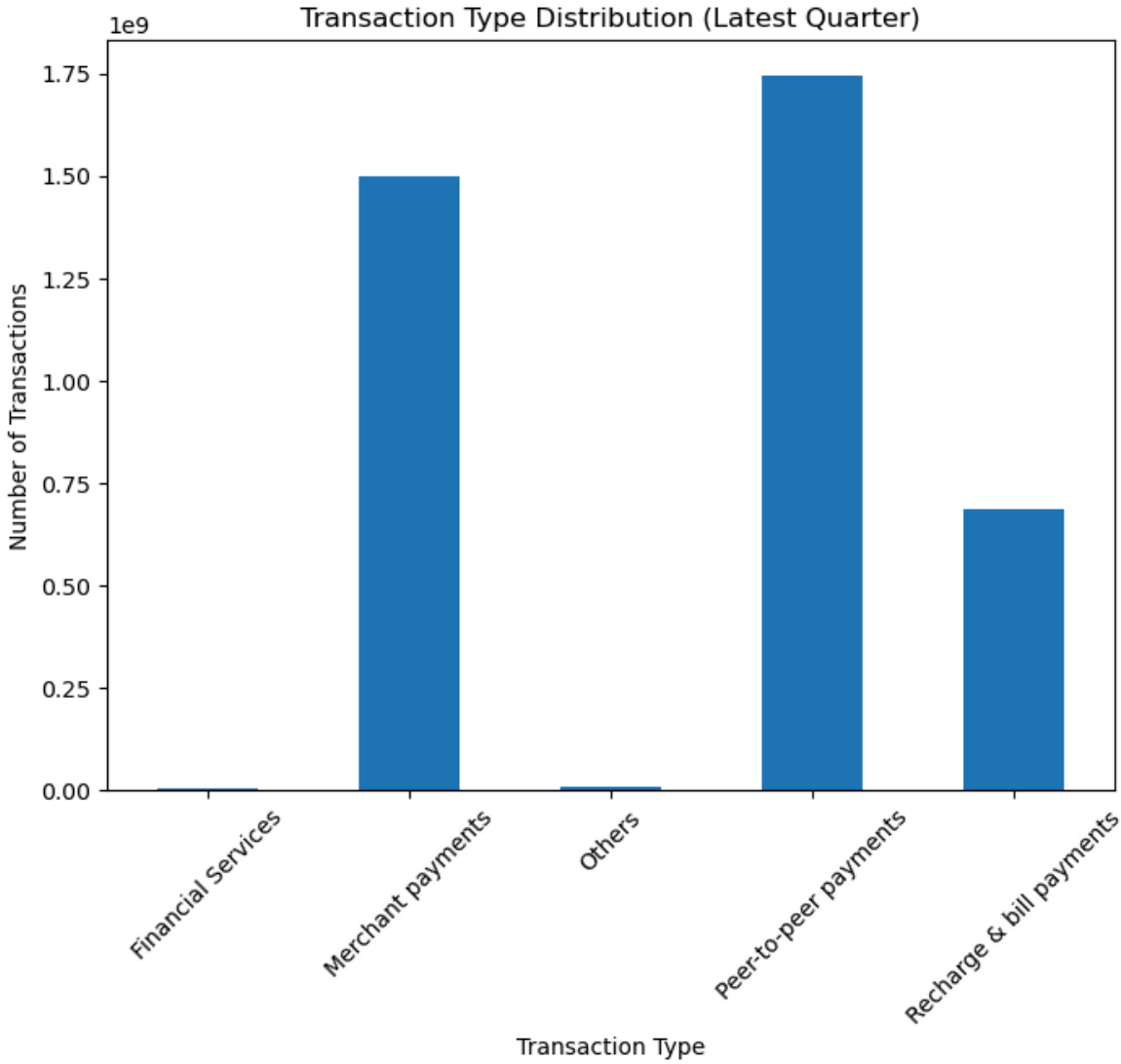
Task 2.7 Distribution of Transaction Types

In [41]:
```
# Find latest year
latest_year = state_txn_split["Year"].max()

# Find latest quarter inside that year
latest_quarter = state_txn_split[state_txn_split["Year"] == latest_year]["Quarte

# Filter data
recent_data = state_txn_split[
    (state_txn_split["Year"] == latest_year) &
    (state_txn_split["Quarter"] == latest_quarter)
]

# Group data
txn_distribution = recent_data.groupby("Transaction Type")["Transactions"].sum()

print(txn_distribution)
```

```
Transaction Type
Financial Services             3397356
Merchant payments           1499624295
Others                         9061701
Peer-to-peer payments       1744265068
Recharge & bill payments     685022045
Name: Transactions, dtype: int64
```

In [42]:
```
plt.figure(figsize=(8,6))

txn_distribution.plot(kind="bar")
```

```
plt.title("Transaction Type Distribution (Latest Quarter)")
plt.xlabel("Transaction Type")
plt.ylabel("Number of Transactions")

plt.xticks(rotation=45)

plt.show()
```



Task 2.8 Unique District-Code Mapping

Here we identify unique mappings between district names and district codes.

In [43]:
```
district_mapping = district_txn_users[["District","Code"]].drop_duplicates()

district_mapping
```

Out[43]:

	District	Code
0	Nicobars	AN01
1	North And Middle Andaman	AN02
2	South Andaman	AN03
42	Anantapur	AP01
43	Chittoor	AP02
44	East Godavari	AP03
45	Guntur	AP04
46	Krishna	AP05
47	Kurnool	AP06
48	Prakasam	AP07
49	Spsr Nellore	AP08
50	Srikakulam	AP09
51	Visakhapatnam	AP10
52	Vizianagaram	AP11
53	West Godavari	AP12
54	Ysr	AP13
224	Anjaw	AR01
225	Changlang	AR02
226	Dibang Valley	AR21
227	East Kameng	AR03
228	East Siang	AR04
229	Kamle	AR05
230	Kra Daadi	AR06
231	Kurung Kumey	AR07
232	Lepa Rada	AR08
233	Lohit	AR09
234	Longding	AR10
235	Lower Dibang Valley	AR11
236	Lower Siang	AR12
237	Lower Subansiri	AR13
238	Namsai	AR14
239	Pakke Kessang	AR15
240	Papum Pare	AR16

	District	Code
241	Shi Yomi	AR17
242	Siang	AR18
243	Tawang	AR19
244	Tirap	AR20
245	Upper Siang	AR22
246	Upper Subansiri	AR23
247	West Kameng	AR24
248	West Siang	AR25
574	Baksa	AS02
575	Barpeta	AS03
576	Biswanath	AS04
577	Bongaigaon	AS05
578	Cachar	AS06
579	Charaideo	AS07
580	Chirang	AS08
581	Darrang	AS09
582	Dhemaji	AS10
583	Dhubri	AS11
584	Dibrugarh	AS12
585	Dima Hasao	AS13
586	Goalpara	AS14
587	Golaghat	AS15
588	Hailakandi	AS16
589	Hojai	AS17
590	Jorhat	AS18
591	Kamrup	AS19
592	Kamrup Metropolitan	AS20
593	Karbi Anglong	AS21
594	Karimganj	AS22
595	Kokrajhar	AS23
596	Lakhimpur	AS24
597	Majuli	AS25
598	Marigaon	AS26

	District	Code
599	Nagaon	AS27
600	Nalbari	AS28
601	Sivasagar	AS29
602	Sonitpur	AS30
603	South Salmara Mancachar	AS31
604	Tinsukia	AS32
605	Udalguri	AS33
606	West Karbi Anglong	AS34
1036	Araria	BR01
1037	Arwal	BR02
1038	Aurangabad	BR03
1039	Banka	BR04
1040	Begusarai	BR05
1041	Bhagalpur	BR06
1042	Bhojpur	BR07
1043	Buxar	BR08
1044	Darbhanga	BR09
1045	Gaya	BR11
1046	Gopalganj	BR12
1047	Jamui	BR13
1048	Jehanabad	BR14
1049	Kaimur Bhabua	BR15
1050	Katihar	BR16
1051	Khagaria	BR17
1052	Kishanganj	BR18
1053	Lakhisarai	BR19
1054	Madhepura	BR20
1055	Madhubani	BR21
1056	Munger	BR22
1057	Muzaffarpur	BR23
1058	Nalanda	BR24
1059	Nawada	BR25
1060	Pashchim Champaran	BR38

	District	Code
1061	Patna	BR26
1062	Purbi Champaran	BR10
1063	Purnia	BR27
1064	Rohtas	BR28
1065	Saharsa	BR29
1066	Samastipur	BR30
1067	Saran	BR31
1068	Sheikhpura	BR32
1069	Sheohar	BR33
1070	Sitamarhi	BR34
1071	Siwan	BR35
1072	Supaul	BR36
1073	Vaishali	BR37
1568	Chandigarh	CH01
1582	Balod	CG01
1583	Baloda Bazar	CG02
1584	Balrampur	CG03
1585	Bastar	CG04
1586	Bemetara	CG05
1587	Bijapur	CG06
1588	Bilaspur	CG07
1589	Dantewada	CG08
1590	Dhamtari	CG09
1591	Durg	CG10
1592	Gariyaband	CG11
1593	Janjgir Champa	CG13
1594	Jashpur	CG14
1595	Kabirdham	CG15
1596	Kanker	CG16
1597	Kondagaon	CG17
1598	Korba	CG18
1599	Korea	CG19
1600	Mahasamund	CG20

	District	Code
1601	Mungeli	CG21
1602	Narayanpur	CG22
1603	Raigarh	CG23
1604	Raipur	CG24
1605	Rajnandgaon	CG25
1606	Sukma	CG26
1607	Surajpur	CG27
1608	Surguja	CG28
1960	Dadra And Nagar Haveli	DD01
1961	Daman	DD02
1962	Diu	DD03
2002	Central	DL01
2003	East	DL02
2004	New Delhi	DL03
2005	North	DL04
2006	North East	DL05
2007	North West	DL06
2008	Shahdara	DL07
2009	South	DL08
2010	South East Delhi	DL09
2011	South West	DL10
2012	West	DL11
2156	North Goa	GA01
2157	South Goa	GA02
2184	Ahmadabad	GJ01
2185	Amreli	GJ02
2186	Anand	GJ03
2187	Aravallis	GJ04
2188	Banas Kantha	GJ05
2189	Bharuch	GJ06
2190	Bhavnagar	GJ07
2191	Botad	GJ08
2192	Chhotaudepur	GJ09

	District	Code
2193	Devbhumi Dwarka	GJ12
2194	Dohad	GJ10
2195	Gandhinagar	GJ13
2196	Gir Somnath	GJ14
2197	Jamnagar	GJ15
2198	Junagadh	GJ16
2199	Kachchh	GJ18
2200	Kheda	GJ17
2201	Mahesana	GJ20
2202	Mahisagar	GJ19
2203	Morbi	GJ21
2204	Narmada	GJ22
2205	Navsari	GJ23
2206	Panch Mahals	GJ24
2207	Patan	GJ25
2208	Porbandar	GJ26
2209	Rajkot	GJ27
2210	Sabar Kantha	GJ28
2211	Surat	GJ29
2212	Surendranagar	GJ30
2213	Tapi	GJ31
2214	The Dangs	GJ11
2215	Vadodara	GJ32
2216	Valsad	GJ33
2646	Ambala	HR01
2647	Bhiwani	HR02
2648	Charkhi Dadri	HR03
2649	Faridabad	HR04
2650	Fatehabad	HR05
2651	Gurugram	HR06
2652	Hisar	HR07
2653	Jhajjar	HR08
2654	Jind	HR09

	District	Code
2655	Kaithal	HR10
2656	Karnal	HR11
2657	Kurukshetra	HR12
2658	Mahendragarh	HR13
2659	Mewat	HR14
2660	Palwal	HR15
2661	Panchkula	HR16
2662	Panipat	HR17
2663	Rewari	HR18
2664	Rohtak	HR19
2665	Sirsa	HR20
2666	Sonipat	HR21
2667	Yamunanagar	HR22
2954	Bilaspur	HP01
2955	Chamba	HP02
2956	Hamirpur	HP03
2957	Kangra	HP04
2958	Kinnaur	HP05
2959	Kullu	HP06
2960	Lahul And Spiti	HP07
2961	Mandi	HP08
2962	Shimla	HP09
2963	Sirmaur	HP10
2964	Solan	HP11
2965	Una	HP12
3122	Anantnag	JK01
3123	Bandipore	JK02
3124	Baramulla	JK03
3125	Budgam	JK04
3126	Doda	JK05
3127	Ganderbal	JK06
3128	Jammu	JK07
3129	Kathua	JK08

	District	Code
3130	Kishtwar	JK09
3131	Kulgam	JK10
3132	Kupwara	JK11
3133	Mirpur	NaN
3134	Muzaffarabad	NaN
3135	Poonch	JK12
3136	Pulwama	JK13
3137	Rajouri	JK14
3138	Ramban	JK15
3139	Reasi	JK16
3140	Samba	JK17
3141	Shopian	JK18
3142	Srinagar	JK19
3143	Udhampur	JK20
3430	Bokaro	JH01
3431	Chatra	JH02
3432	Deoghar	JH03
3433	Dhanbad	JH04
3434	Dumka	JH05
3435	East Singhbhum	JH06
3436	Garhwa	JH07
3437	Giridih	JH08
3438	Godda	JH09
3439	Gumla	JH10
3440	Hazaribagh	JH11
3441	Jamtara	JH12
3442	Khunti	JH13
3443	Koderma	JH14
3444	Latehar	JH15
3445	Lohardaga	JH16
3446	Pakur	JH17
3447	Palamu	JH18
3448	Ramgarh	JH19

	District	Code
3449	Ranchi	JH20
3450	Sahebganj	JH21
3451	Saraikela Kharsawan	JH22
3452	Simdega	JH23
3453	West Singhbhum	JH24
3766	Bagalkote	KA01
3767	Ballari	KA02
3768	Belagavi	KA05
3769	Bengaluru Rural	KA03
3770	Bengaluru Urban	KA04
3771	Bidar	KA06
3772	Chamarajanagara	KA08
3773	Chikkaballapura	KA09
3774	Chikkamagaluru	KA10
3775	Chitradurga	KA11
3776	Dakshina Kannada	KA12
3777	Davanagere	KA13
3778	Dharwad	KA14
3779	Gadag	KA15
3780	Hassan	KA17
3781	Haveri	KA18
3782	Kalaburagi	KA16
3783	Kodagu	KA19
3784	Kolar	KA20
3785	Koppal	KA21
3786	Mandya	KA22
3787	Mysuru	KA23
3788	Raichur	KA24
3789	Ramanagara	KA25
3790	Shivamogga	KA26
3791	Tumakuru	KA27
3792	Udupi	KA28
3793	Uttara Kannada	KA29

	District	Code
3794	Vijayapura	KA07
3795	Yadgir	KA31
4186	Alappuzha	KL01
4187	Ernakulam	KL02
4188	Idukki	KL03
4189	Kannur	KL04
4190	Kasaragod	KL05
4191	Kollam	KL06
4192	Kottayam	KL07
4193	Kozhikode	KL08
4194	Malappuram	KL09
4195	Palakkad	KL10
4196	Pathanamthitta	KL11
4197	Thiruvananthapuram	KL12
4198	Thrissur	KL13
4199	Wayanad	KL14
4382	Kargil	LA01
4383	Leh Ladakh	LA02
4410	Lakshadweep	LD01
4424	Agar Malwa	MP01
4425	Alirajpur	MP02
4426	Anuppur	MP03
4427	Ashoknagar	MP04
4428	Balaghat	MP05
4429	Barwani	MP06
4430	Betul	MP07
4431	Bhind	MP08
4432	Bhopal	MP09
4433	Burhanpur	MP10
4434	Chhatarpur	MP12
4435	Chhindwara	MP13
4436	Damoh	MP14
4437	Datia	MP15

	District	Code
4438	Dewas	MP16
4439	Dhar	MP17
4440	Dindori	MP18
4441	East Nimar	MP27
4442	Guna	MP19
4443	Gwalior	MP20
4444	Harda	MP21
4445	Hoshangabad	MP22
4446	Indore	MP23
4447	Jabalpur	MP24
4448	Jhabua	MP25
4449	Katni	MP26
4450	Khargone	MP28
4451	Mandla	MP30
4452	Mandsaur	MP31
4453	Morena	MP32
4454	Narsinghpur	MP34
4455	Neemuch	MP35
4456	Niwari	MP36
4457	Panna	MP37
4458	Raisen	MP38
4459	Rajgarh	MP39
4460	Ratlam	MP40
4461	Rewa	MP41
4462	Sagar	MP42
4463	Satna	MP43
4464	Sehore	MP44
4465	Seoni	MP45
4466	Shahdol	MP46
4467	Shajapur	MP47
4468	Sheopur	MP48
4469	Shivpuri	MP49
4470	Sidhi	MP50

	District	Code
4471	Singrauli	MP51
4472	Tikamgarh	MP52
4473	Ujjain	MP53
4474	Umaria	MP54
4475	Vidisha	MP55
5152	Ahmednagar	MH01
5153	Akola	MH02
5154	Amravati	MH03
5155	Aurangabad	MH04
5156	Beed	MH05
5157	Bhandara	MH06
5158	Buldhana	MH07
5159	Chandrapur	MH08
5160	Dhule	MH09
5161	Gadchiroli	MH10
5162	Gondia	MH11
5163	Hingoli	MH12
5164	Jalgaon	MH13
5165	Jalna	MH14
5166	Kolhapur	MH15
5167	Latur	MH16
5168	Mumbai	MH17
5169	Mumbai Suburban	MH18
5170	Nagpur	MH19
5171	Nanded	MH20
5172	Nandurbar	MH21
5173	Nashik	MH22
5174	Osmanabad	MH23
5175	Palghar	MH24
5176	Parbhani	MH25
5177	Pune	MH26
5178	Raigad	MH27
5179	Ratnagiri	MH28

	District	Code
5180	Sangli	MH29
5181	Satara	MH30
5182	Sindhudurg	MH31
5183	Solapur	MH32
5184	Thane	MH33
5185	Wardha	MH34
5186	Washim	MH35
5187	Yavatmal	MH36
5656	Bishnupur	MN01
5657	Chandel	MN02
5658	Churachandpur	MN03
5659	Imphal East	MN04
5660	Imphal West	MN05
5661	Jiribam	MN06
5662	Kakching	MN07
5663	Kamjong	MN08
5664	Kangpokpi	MN09
5665	Noney	MN10
5666	Pherzawl	MN11
5667	Senapati	MN12
5668	Tamenglong	MN13
5669	Tengnoupal	MN14
5670	Thoubal	MN15
5671	Ukhrul	MN16
5880	East Garo Hills	ML01
5881	East Jaintia Hills	ML02
5882	East Khasi Hills	ML03
5883	North Garo Hills	ML04
5884	Ribhoi	ML05
5885	South Garo Hills	ML06
5886	South West Garo Hills	ML07
5887	South West Khasi Hills	ML08
5888	West Garo Hills	ML09

	District	Code
5889	West Jaintia Hills	ML10
5890	West Khasi Hills	ML11
6034	Aizawl	MZ01
6035	Champhai	MZ02
6036	Kolasib	MZ05
6037	Lawngtlai	MZ06
6038	Lunglei	MZ07
6039	Mamit	MZ08
6040	Saiha	MZ09
6041	Serchhip	MZ11
6146	Dimapur	NL01
6147	Kiphire	NL02
6148	Kohima	NL03
6149	Longleng	NL04
6150	Mokokchung	NL05
6151	Mon	NL06
6152	Peren	NL08
6153	Phek	NL09
6154	Tuensang	NL10
6155	Wokha	NL11
6156	Zunheboto	NL12
6300	Anugul	OR01
6301	Balangir	OR02
6302	Baleshwar	OR03
6303	Bargarh	OR04
6304	Bhadrak	OR05
6305	Boudh	OR06
6306	Cuttack	OR07
6307	Deogarh	OR08
6308	Dhenkanal	OR09
6309	Gajapati	OR10
6310	Ganjam	OR11
6311	Jagatsinghapur	OR12

	District	Code
6312	Jajapur	OR13
6313	Jharsuguda	OR14
6314	Kalahandi	OR15
6315	Kandhamal	OR16
6316	Kendrapara	OR17
6317	Kendujhar	OR18
6318	Khordha	OR19
6319	Koraput	OR20
6320	Malkangiri	OR21
6321	Mayurbhanj	OR22
6322	Nabarangpur	OR23
6323	Nayagarh	OR24
6324	Nuapada	OR25
6325	Puri	OR26
6326	Rayagada	OR27
6327	Sambalpur	OR28
6328	Sonepur	OR29
6329	Sundargarh	OR30
6720	Karaikal	PY01
6721	Mahe	PY02
6722	Puducherry	PY03
6723	Yanam	PY04
6776	Amritsar	PB01
6777	Barnala	PB02
6778	Bathinda	PB03
6779	Faridkot	PB04
6780	Fatehgarh Sahib	PB05
6781	Fazilka	PB06
6782	Firozepur	PB07
6783	Gurdaspur	PB08
6784	Hoshiarpur	PB09
6785	Jalandhar	PB10
6786	Kapurthala	PB11

	District	Code
6787	Ludhiana	PB12
6788	Mansa	PB14
6789	Moga	PB15
6790	Pathankot	PB16
6791	Patiala	PB17
6792	Rupnagar	PB18
6793	Sangrur	PB20
6794	Sas Nagar	PB19
6795	Shahid Bhagat Singh Nagar	PB21
6796	Sri Muktsar Sahib	PB22
6797	Tarn Taran	PB23
7084	Ajmer	RJ01
7085	Alwar	RJ02
7086	Banswara	RJ03
7087	Baran	RJ04
7088	Barmer	RJ05
7089	Bharatpur	RJ06
7090	Bhilwara	RJ07
7091	Bikaner	RJ08
7092	Bundi	RJ09
7093	Chittorgarh	RJ10
7094	Churu	RJ11
7095	Dausa	RJ12
7096	Dholpur	RJ13
7097	Dungarpur	RJ14
7098	Ganganagar	RJ31
7099	Hanumangarh	RJ15
7100	Jaipur	RJ16
7101	Jaisalmer	RJ17
7102	Jalore	RJ18
7103	Jhalawar	RJ19
7104	Jhunjhunu	RJ20
7105	Jodhpur	RJ21

	District	Code
7106	Karauli	RJ22
7107	Kota	RJ23
7108	Nagaur	RJ24
7109	Pali	RJ25
7110	Pratapgarh	RJ26
7111	Rajsamand	RJ27
7112	Sawai Madhopur	RJ28
7113	Sikar	RJ29
7114	Sirohi	RJ30
7115	Tonk	RJ32
7116	Udaipur	RJ33
7546	East	SK01
7547	North	SK02
7548	South	SK03
7549	West	SK04
7602	Ariyalur	TN01
7603	Chengalpattu	TN02
7604	Chennai	TN03
7605	Coimbatore	TN04
7606	Cuddalore	TN05
7607	Dharmapuri	TN06
7608	Dindigul	TN07
7609	Erode	TN08
7610	Kallakkurichi	TN09
7611	Kancheepuram	TN10
7612	Kanniyakumari	TN11
7613	Karur	TN12
7614	Krishnagiri	TN13
7615	Madurai	TN14
7616	Nagapattinam	TN16
7617	Namakkal	TN17
7618	Perambalur	TN19
7619	Pudukkottai	TN20

	District	Code
7620	Ramanathapuram	TN21
7621	Ranipet	TN22
7622	Salem	TN23
7623	Sivaganga	TN24
7624	Tenkasi	TN25
7625	Thanjavur	TN26
7626	The Nilgiris	TN18
7627	Theni	TN27
7628	Thiruvallur	TN33
7629	Thiruvarur	TN35
7630	Thoothukkudi	TN28
7631	Tiruchirappalli	TN29
7632	Tirunelveli	TN30
7633	Tirupathur	TN31
7634	Tiruppur	TN32
7635	Tiruvannamalai	TN34
7636	Vellore	TN36
7637	Viluppuram	TN37
7638	Virudhunagar	TN38
8120	Adilabad	TG01
8121	Bhadradri Kothagudem	TG02
8122	Hyderabad	TG03
8123	Jagtial	TG04
8124	Jangaon	TG05
8125	Jaya Shankar Bhalupally	TG06
8126	Jogulamba Gadwal	TG07
8127	Kamareddy	TG08
8128	Karimnagar	TG09
8129	Khammam	TG10
8130	Kumuram Bheem Asifabad	TG11
8131	Mahabubabad	TG12
8132	Mahbubnagar	TG13
8133	Mancherial	TG14

	District	Code
8134	Medak	TG15
8135	Medchal Malkajgiri	TG16
8136	Mulugu	TG17
8137	Nagarkurnool	TG18
8138	Nalgonda	TG19
8139	Narayanpet	TG20
8140	Nirmal	TG21
8141	Nizamabad	TG22
8142	Peddapalle	TG23
8143	Rajanna Sircilla	TG24
8144	Rangareddy	TG25
8145	Sangareddy	TG26
8146	Siddipet	TG27
8147	Suryapet	TG28
8148	Vikarabad	TG29
8149	Wanaparthy	TG30
8150	Warangal Rural	TG31
8151	Warangal Urban	TG32
8152	Yadadri Bhuvanagiri	TG33
8582	Dhalai	TR01
8583	Gomati	TR02
8584	Khowai	TR03
8585	North Tripura	TR04
8586	Sepahijala	TR05
8587	South Tripura	TR06
8588	Unakoti	TR07
8589	West Tripura	TR08
8694	Almora	UK01
8695	Bageshwar	UK02
8696	Chamoli	UK03
8697	Champawat	UK04
8698	Dehradun	UK05
8699	Haridwar	UK06

	District	Code
8700	Nainital	UK07
8701	Pauri Garhwal	UK08
8702	Pithoragarh	UK09
8703	Rudraprayag	UK10
8704	Tehri Garhwal	UK11
8705	Udham Singh Nagar	UK12
8706	Uttarkashi	UK13
8876	Agra	UP01
8877	Aligarh	UP02
8878	Ambedkar Nagar	UP04
8879	Amethi	UP05
8880	Amroha	UP06
8881	Auraiya	UP07
8882	Ayodhya	UP26
8883	Azamgarh	UP08
8884	Baghpat	UP09
8885	Bahraich	UP10
8886	Ballia	UP11
8887	Balrampur	UP12
8888	Banda	UP13
8889	Bara Banki	UP14
8890	Bareilly	UP15
8891	Basti	UP16
8892	Bhadohi	UP17
8893	Bijnor	UP18
8894	Budaun	UP19
8895	Bulandshahr	UP20
8896	Chandauli	UP21
8897	Chitrakoot	UP22
8898	Deoria	UP23
8899	Etah	UP24
8900	Etawah	UP25
8901	Farrukhabad	UP27

	District	Code
8902	Fatehpur	UP28
8903	Firozabad	UP29
8904	Gautam Buddha Nagar	UP30
8905	Ghaziabad	UP31
8906	Ghazipur	UP32
8907	Gonda	UP33
8908	Gorakhpur	UP34
8909	Hamirpur	UP35
8910	Hapur	UP36
8911	Hardoi	UP37
8912	Hathras	UP38
8913	Jalaun	UP39
8914	Jaunpur	UP40
8915	Jhansi	UP41
8916	Kannauj	UP42
8917	Kanpur Dehat	UP43
8918	Kanpur Nagar	UP44
8919	Kasganj	UP45
8920	Kaushambi	UP46
8921	Kheri	UP48
8922	Kushinagar	UP47
8923	Lalitpur	UP49
8924	Lucknow	UP50
8925	Maharajganj	UP51
8926	Mahoba	UP52
8927	Mainpuri	UP53
8928	Mathura	UP54
8929	Mau	UP55
8930	Meerut	UP56
8931	Mirzapur	UP57
8932	Moradabad	UP58
8933	Muzaffarnagar	UP59
8934	Pilibhit	UP60

	District	Code
8935	Pratapgarh	UP61
8936	Prayagraj	UP03
8937	Rae Bareli	UP62
8938	Rampur	UP63
8939	Saharanpur	UP64
8940	Sambhal	UP65
8941	Sant Kabeer Nagar	UP66
8942	Shahjahanpur	UP67
8943	Shamli	UP68
8944	Shravasti	UP69
8945	Siddharthnagar	UP70
8946	Sitapur	UP71
8947	Sonbhadra	UP72
8948	Sultanpur	UP73
8949	Unnao	UP74
8950	Varanasi	UP75
9926	Alipurduar	WB01
9927	Bankura	WB02
9928	Birbhum	WB03
9929	Dakshin Dinajpur	WB05
9930	Darjiling	WB06
9931	Hooghly	WB07
9932	Howrah	WB08
9933	Jalpaiguri	WB09
9934	Jhargram	WB10
9935	Kalimpong	WB11
9936	Koch Bihar	WB04
9937	Kolkata	WB12
9938	Maldah	WB13
9939	Murshidabad	WB14
9940	Nadia	WB15
9941	North Twenty Four Parganas	WB16
9942	Paschim Bardhaman	WB17

	District	Code
9943	Paschim Medinipur	WB18
9944	Purba Bardhaman	WB19
9945	Purba Medinipur	WB20
9946	Purulia	WB21
9947	South Twenty Four Parganas	WB22
9948	Uttar Dinajpur	WB23

In [44]:
```
district_mapping.to_csv("district_code_mapping.csv", index=False)
```

Interpretation

A clean mapping between district names and district codes was created and exported as a CSV file for easier reference and data integration.

Task 2 Summary

During the Exploratory Data Analysis stage, we discovered several insights:

• Specific smartphone brands are widely used across states.

• App usage trends indicate growing engagement over time.

These insights help understand the digital payment ecosystem in India and prepare the data for deeper analysis in subsequent tasks.

Task 3: Data Quality Checks

In this section, we validate the consistency of the dataset by comparing aggregated district-level data with the corresponding state-level data.

The objective is to ensure that:

• District level totals match state level totals
• There are no discrepancies in transactions, amounts, or registered users

This step is essential in real-world analytics because data often originates from multiple sources.

Task 3.1 Validate State and District Data Consistency

In [47]:
```
district_state_agg = district_txn_users.groupby("State")[[
    "Transactions",
```

```
    "Amount (INR)",
    "Registered Users"
]].sum().reset_index()

district_state_agg.head()
```

Out[47]:

	State	Transactions	Amount (INR)	Registered Users
0	Andaman & Nicobar Islands	1223565	3.641818e+09	408396
1	Andhra Pradesh	1781091169	3.860292e+12	158007957
2	Arunachal Pradesh	8043799	2.225183e+10	1862214
3	Assam	123577981	2.814087e+11	26860751
4	Bihar	713643889	1.639579e+12	115284760

The district-level dataset is grouped by state and aggregated to calculate:

- Total transactions
- Total transaction amount
- Total registered users

In [48]:

```
state_level_agg = state_txn_users.groupby("State")[[
    "Transactions",
    "Amount (INR)",
    "Registered Users"
]].sum().reset_index()

state_level_agg.head()
```

Out[48]:

	State	Transactions	Amount (INR)	Registered Users
0	Andaman & Nicobar Islands	1223565	3.641818e+09	408396
1	Andhra Pradesh	1781091169	3.187847e+12	158007957
2	Arunachal Pradesh	8043799	2.225183e+10	1862214
3	Assam	123577981	2.814087e+11	26860751
4	Bihar	713643889	1.639579e+12	115284760

In [49]:

```
comparison = pd.merge(
    state_level_agg,
    district_state_agg,
    on="State",
    suffixes=("_State","_District")
)

comparison.head()
```

Out[49]:

	State	Transactions_State	Amount (INR)_State	Registered Users_State	Transactions_District	A (INR)_
0	Andaman & Nicobar Islands	1223565	3.641818e+09	408396	1223565	3.6418
1	Andhra Pradesh	1781091169	3.187847e+12	158007957	1781091169	3.8602
2	Arunachal Pradesh	8043799	2.225183e+10	1862214	8043799	2.2251
3	Assam	123577981	2.814087e+11	26860751	123577981	2.8140
4	Bihar	713643889	1.639579e+12	115284760	713643889	1.6395

In [50]:

```
discrepancies = comparison[
    (comparison["Transactions_State"] != comparison["Transactions_District"]) |
    (comparison["Amount (INR)_State"] != comparison["Amount (INR)_District"]) |
    (comparison["Registered Users_State"] != comparison["Registered Users_Distri
]

discrepancies
```

Out[50]:

	State	Transactions_State	Amount (INR)_State	Registered Users_State	Transactions_District	(I
0	Andaman & Nicobar Islands	1223565	3.641818e+09	408396	1223565	3.
1	Andhra Pradesh	1781091169	3.187847e+12	158007957	1781091169	3.
2	Arunachal Pradesh	8043799	2.225183e+10	1862214	8043799	2.
3	Assam	123577981	2.814087e+11	26860751	123577981	2.
4	Bihar	713643889	1.639579e+12	115284760	713643889	1.
5	Chandigarh	25516542	5.371208e+10	3801466	25516542	5.
6	Chhattisgarh	241908549	4.156353e+11	37448388	241908549	4.
7	Dadra & Nagar Haveli and Daman & Diu	12462185	1.865878e+10	2701112	12462185	1.
8	Delhi	1011031124	1.649942e+12	95374654	1011031124	1.
9	Goa	24481604	4.427187e+10	4755320	24481604	4.
10	Gujarat	612992823	1.105329e+12	126380032	612992823	1.
11	Haryana	567751410	1.059832e+12	85564119	567751410	1.
12	Himachal Pradesh	52987445	9.786546e+10	13373980	52987445	9.
13	Jammu & Kashmir	45053328	1.094954e+11	9756438	45053328	1.
14	Jharkhand	275270987	4.905701e+11	44768682	275270987	4.
15	Karnataka	2981044533	4.361396e+12	203098564	2981044533	4.
16	Kerala	223637876	3.729599e+11	51267042	223637876	3.
17	Ladakh	1880109	6.606978e+09	579825	1880109	6.
18	Lakshadweep	71610	1.921587e+08	34223	71610	1.
19	Madhya Pradesh	1100253728	1.915101e+12	125339823	1100253728	1.
20	Maharashtra	2833670154	4.290995e+12	315783658	2833670154	4.
22	Meghalaya	5648913	1.254162e+10	1436472	5648913	1.
23	Mizoram	2162776	6.317141e+09	605612	2162776	6.
24	Nagaland	5864977	1.638757e+10	1440387	5864977	1.
25	Odisha	740479146	1.220615e+12	80404103	740479146	1.
26	Puducherry	17292386	3.209769e+10	2785162	17292386	3.

	State	Transactions_State	Amount (INR)_State	Registered Users_State	Transactions_District	(I
27	Punjab	181336904	3.941773e+11	45811761	181336904	3.
28	Rajasthan	1382918930	2.625584e+12	149770064	1382918930	2.
29	Sikkim	6483948	1.415554e+10	1279199	6483948	1.
30	Tamil Nadu	800399873	1.481448e+12	132437426	800399873	1.
31	Telangana	2347430243	4.574246e+12	149224214	2347430243	4.
32	Tripura	12018152	2.097051e+10	3661575	12018152	2.
33	Uttar Pradesh	1314714390	2.399757e+12	246200404	1314714390	2.
34	Uttarakhand	124011162	2.150715e+11	23792747	124011162	2.
35	West Bengal	942286041	1.452707e+12	144242183	942286041	1.

Interpretation

If the discrepancies table is empty, it indicates that the district-level totals perfectly match the state-level totals.

If differences exist, it suggests potential issues such as:

- Missing district records
- Aggregation inconsistencies
- Data synchronization problems

Such validation checks are commonly performed in real-world data pipelines.

Task 4: Data Merging and Advanced Analysis

In this section, we combine multiple datasets to perform deeper analysis.

The analysis includes:

- Ratio of registered users to population
- Correlation between population density and transaction volume
- Average transaction amount per user
- Device brand usage ratios

Task 4.1 User to Population Ratio

We merge demographic data with state-level user data to calculate the ratio of registered users to the population.

In [52]:
```
state_population = district_demographics.groupby("State")["Population"].sum().re

state_population.head()
```

Out[52]:

	State	Population
0	Andaman & Nicobar Islands	380581
1	Andhra Pradesh	49378776
2	Arunachal Pradesh	1606047
3	Assam	34461385
4	Bihar	103817951

In [53]:
```
state_users = state_txn_users.groupby("State")["Registered Users"].max().reset_i

state_user_population = pd.merge(
    state_users,
    state_population,
    on="State"
)

state_user_population.head()
```

Out[53]:

	State	Registered Users	Population
0	Andaman & Nicobar Islands	62095	380581
1	Andhra Pradesh	19620905	49378776
2	Arunachal Pradesh	270572	1606047
3	Assam	3617056	34461385
4	Bihar	14972134	103817951

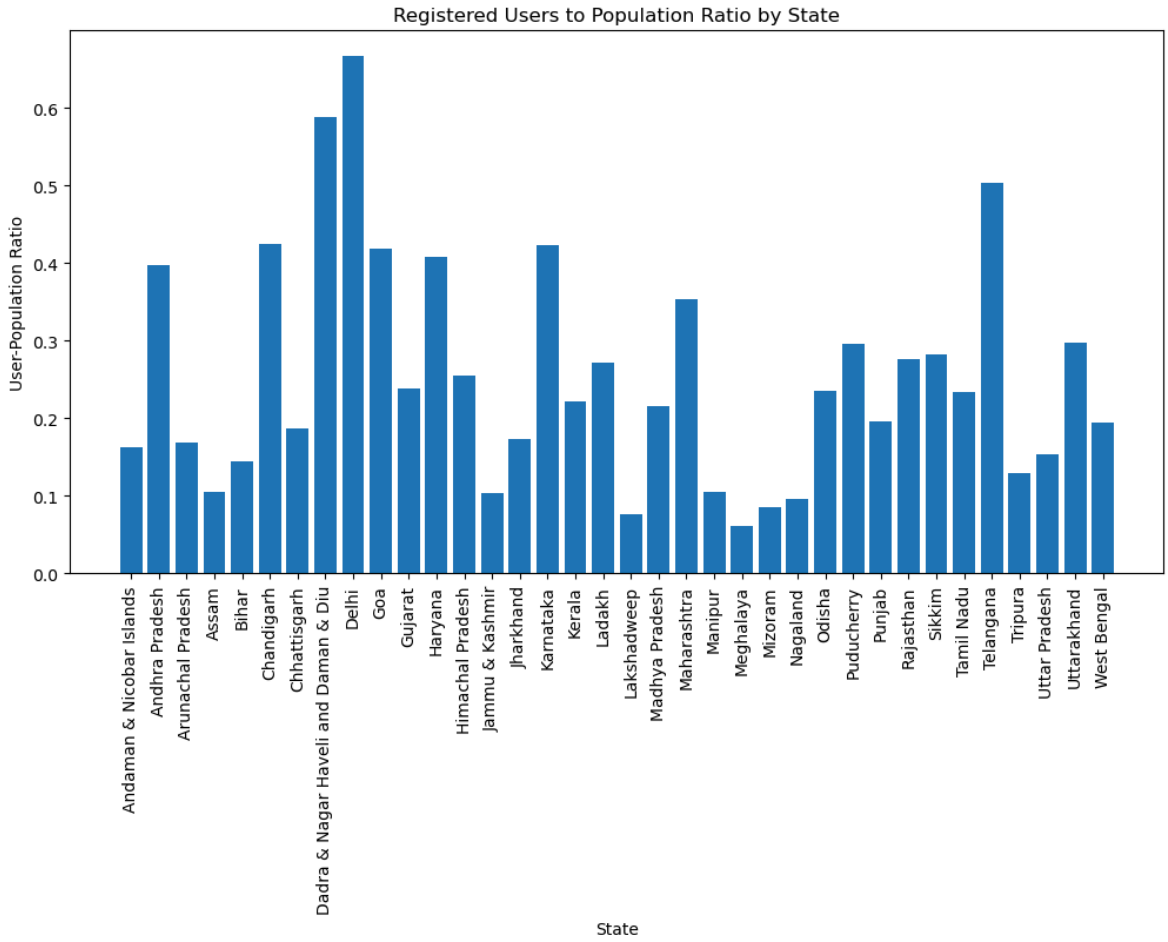
In [54]:
```
state_user_population["User_Population_Ratio"] = (
    state_user_population["Registered Users"] /
    state_user_population["Population"]
)

state_user_population
```

Out[54]:

	State	Registered Users	Population	User_Population_Ratio
0	Andaman & Nicobar Islands	62095	380581	0.163158
1	Andhra Pradesh	19620905	49378776	0.397355
2	Arunachal Pradesh	270572	1606047	0.168471
3	Assam	3617056	34461385	0.104960
4	Bihar	14972134	103817951	0.144215
5	Chandigarh	448572	1055450	0.425005
6	Chhattisgarh	4837055	25867958	0.186990
7	Dadra & Nagar Haveli and Daman & Diu	344994	586956	0.587768
8	Delhi	11182699	16787941	0.666115
9	Goa	609106	1457723	0.417848
10	Gujarat	15735981	65930328	0.238676
11	Haryana	10530594	25855357	0.407289
12	Himachal Pradesh	1743151	6856509	0.254233
13	Jammu & Kashmir	1261267	12258093	0.102893
14	Jharkhand	5701503	32966238	0.172950
15	Karnataka	25751023	60965960	0.422384
16	Kerala	7416083	33387677	0.222120
17	Ladakh	75048	276875	0.271054
18	Lakshadweep	4896	64473	0.075939
19	Madhya Pradesh	15940935	73943914	0.215581
20	Maharashtra	39664697	112378989	0.352955
21	Manipur	335409	3187465	0.105228
22	Meghalaya	205686	3364915	0.061127
23	Mizoram	92552	1091014	0.084831
24	Nagaland	223337	2335175	0.095640
25	Odisha	9891029	41947358	0.235796
26	Puducherry	369961	1247953	0.296454
27	Punjab	5976548	30510873	0.195883
28	Rajasthan	18980985	68620629	0.276608
29	Sikkim	171740	607688	0.282612
30	Tamil Nadu	17804542	76018492	0.234213
31	Telangana	18306880	36366570	0.503399

	State	Registered Users	Population	User_Population_Ratio
32	Tripura	472153	3671032	0.128616
33	Uttar Pradesh	31651894	205867440	0.153749
34	Uttarakhand	3004040	10116752	0.296937
35	West Bengal	17979979	92828987	0.193689

In [55]:
```
plt.figure(figsize=(12,6))

plt.bar(
    state_user_population["State"],
    state_user_population["User_Population_Ratio"]
)

plt.xticks(rotation=90)

plt.title("Registered Users to Population Ratio by State")

plt.xlabel("State")

plt.ylabel("User-Population Ratio")

plt.show()
```



States with higher user-population ratios indicate stronger adoption of digital payment platforms.

Lower ratios may suggest opportunities for increased digital payment penetration.

Task 4.2 Population Density vs Transaction Volume

This analysis investigates whether districts with higher population density also exhibit higher transaction volumes.

In [57]:
```
district_merge = pd.merge(
    district_txn_users,
    district_demographics,
    on=["State","District","Code"]
)

district_merge.head()
```

Out[57]:

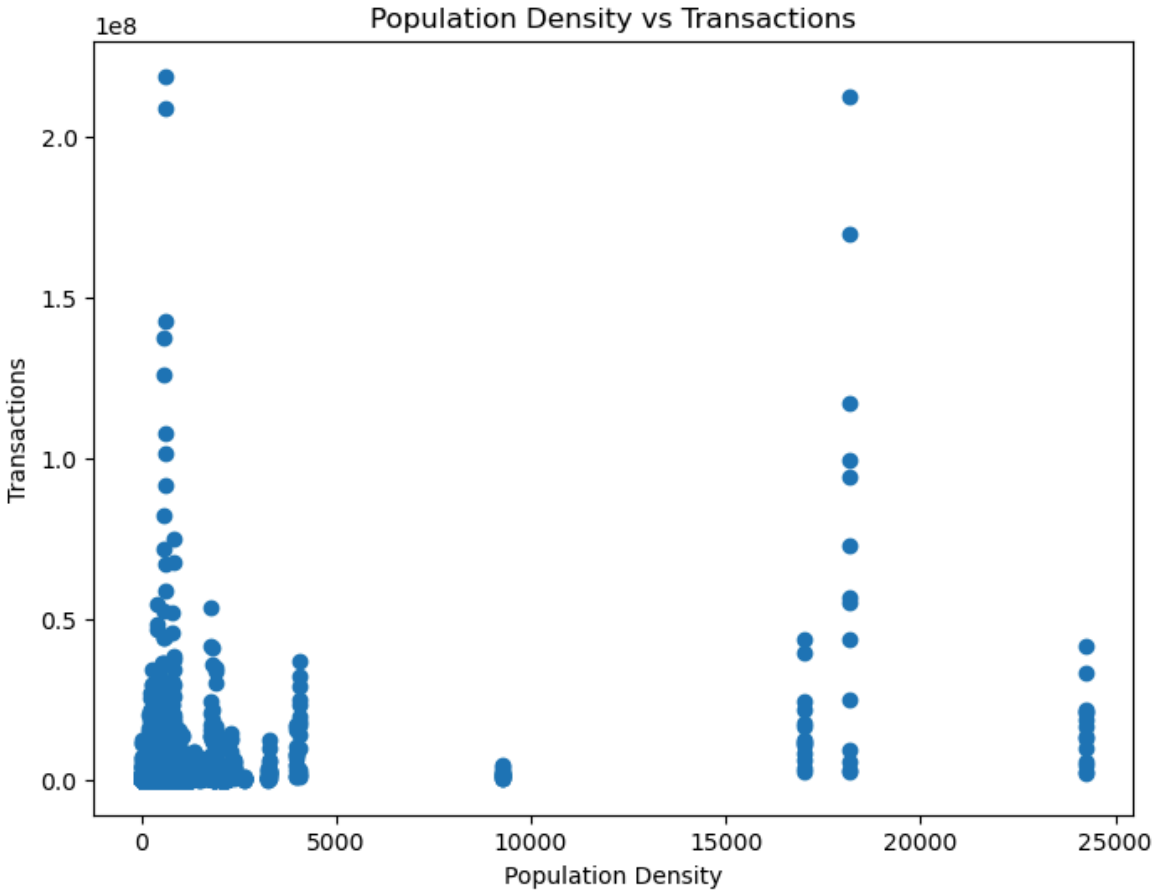
	State	Year	Quarter	District	Code	Transactions	Amount (INR)	ATV (INR)
0	Andaman & Nicobar Islands	2018	1	South Andaman	AN03	5688	1.256025e+07	2208.201361
1	Andaman & Nicobar Islands	2018	2	South Andaman	AN03	9395	2.394824e+07	2549.040502
2	Andaman & Nicobar Islands	2018	3	South Andaman	AN03	13511	4.426811e+07	3276.449742
3	Andaman & Nicobar Islands	2018	4	South Andaman	AN03	20136	7.541184e+07	3745.125189
4	Andaman & Nicobar Islands	2019	1	South Andaman	AN03	25683	8.550197e+07	3329.127142

In [58]:
```
correlation = district_merge["Density"].corr(district_merge["Transactions"])

print("Correlation between Population Density and Transactions:", correlation)
```

```
Correlation between Population Density and Transactions: 0.3077041087559286
```

In [59]:
```
plt.figure(figsize=(8,6))

plt.scatter(
    district_merge["Density"],
    district_merge["Transactions"]
```

```
)

plt.title("Population Density vs Transactions")

plt.xlabel("Population Density")

plt.ylabel("Transactions")

plt.show()
```



The scatter plot helps visualize the relationship between population density and transaction activity.

A positive correlation would indicate that densely populated districts tend to generate more digital payment transactions.

Task 4.3 Average Transaction Amount per User

This metric helps evaluate spending behavior of users across states.

In [61]:
```
state_summary = state_txn_users.groupby("State")[[
    "Amount (INR)",
    "Registered Users"
]].sum().reset_index()

state_summary["Avg_Amount_Per_User"] = (
    state_summary["Amount (INR)"] /
    state_summary["Registered Users"]
)
```

```
state_summary
```

Out[61]:

	State	Amount (INR)	Registered Users	Avg_Amount_Per_User
0	Andaman & Nicobar Islands	3.641818e+09	408396	8917.369075
1	Andhra Pradesh	3.187847e+12	158007957	20175.233763
2	Arunachal Pradesh	2.225183e+10	1862214	11949.126662
3	Assam	2.814087e+11	26860751	10476.576498
4	Bihar	1.639579e+12	115284760	14221.991933
5	Chandigarh	5.371208e+10	3801466	14129.308175
6	Chhattisgarh	4.156353e+11	37448388	11098.884128
7	Dadra & Nagar Haveli and Daman & Diu	1.865878e+10	2701112	6907.815954
8	Delhi	1.649942e+12	95374654	17299.587909
9	Goa	4.427187e+10	4755320	9309.966756
10	Gujarat	1.105329e+12	126380032	8746.070597
11	Haryana	1.059832e+12	85564119	12386.401889
12	Himachal Pradesh	9.786546e+10	13373980	7317.601442
13	Jammu & Kashmir	1.094954e+11	9756438	11222.890558
14	Jharkhand	4.905701e+11	44768682	10957.886396
15	Karnataka	4.361396e+12	203098564	21474.282206
16	Kerala	3.729599e+11	51267042	7274.846690
17	Ladakh	6.606978e+09	579825	11394.780236
18	Lakshadweep	1.921587e+08	34223	5614.898889
19	Madhya Pradesh	1.915101e+12	125339823	15279.269764
20	Maharashtra	4.290995e+12	315783658	13588.401884
21	Manipur	3.579325e+10	2317734	15443.209245
22	Meghalaya	1.254162e+10	1436472	8730.850232
23	Mizoram	6.317141e+09	605612	10431.004396
24	Nagaland	1.638757e+10	1440387	11377.201182
25	Odisha	1.220615e+12	80404103	15180.998375
26	Puducherry	3.209769e+10	2785162	11524.533443
27	Punjab	3.941773e+11	45811761	8604.281444
28	Rajasthan	2.625584e+12	149770064	17530.768707
29	Sikkim	1.415554e+10	1279199	11065.943952
30	Tamil Nadu	1.481448e+12	132437426	11186.022629
31	Telangana	4.574246e+12	149224214	30653.509622

	State	Amount (INR)	Registered Users	Avg_Amount_Per_User
32	Tripura	2.097051e+10	3661575	5727.184275
33	Uttar Pradesh	2.399757e+12	246200404	9747.169644
34	Uttarakhand	2.150715e+11	23792747	9039.371194
35	West Bengal	1.452707e+12	144242183	10071.303655

In [62]:
```
top5_spending_states = state_summary.sort_values(
    by="Avg_Amount_Per_User",
    ascending=False
).head(5)

top5_spending_states
```

Out[62]:

	State	Amount (INR)	Registered Users	Avg_Amount_Per_User
31	Telangana	4.574246e+12	149224214	30653.509622
15	Karnataka	4.361396e+12	203098564	21474.282206
1	Andhra Pradesh	3.187847e+12	158007957	20175.233763
28	Rajasthan	2.625584e+12	149770064	17530.768707
8	Delhi	1.649942e+12	95374654	17299.587909

In [63]:
```
bottom5_spending_states = state_summary.sort_values(
    by="Avg_Amount_Per_User"
).head(5)

bottom5_spending_states
```

Out[63]:

	State	Amount (INR)	Registered Users	Avg_Amount_Per_User
18	Lakshadweep	1.921587e+08	34223	5614.898889
32	Tripura	2.097051e+10	3661575	5727.184275
7	Dadra & Nagar Haveli and Daman & Diu	1.865878e+10	2701112	6907.815954
16	Kerala	3.729599e+11	51267042	7274.846690
12	Himachal Pradesh	9.786546e+10	13373980	7317.601442

Task 4.4 Device Brand Usage Ratio

We analyze the share of different device brands used by PhonePe users in each state.

In [64]:
```
device_merge = pd.merge(
    state_device_data,
    state_users,
    on="State"
```

```
)

device_merge.head()
```

Out[64]:

	State	Year	Quarter	Brand	Registered Users_x	Percentage	Registered Users_y
0	Andaman & Nicobar Islands	2018	1	Xiaomi	1665	0.247033	62095
1	Andaman & Nicobar Islands	2018	1	Samsung	1445	0.214392	62095
2	Andaman & Nicobar Islands	2018	1	Vivo	982	0.145697	62095
3	Andaman & Nicobar Islands	2018	1	Oppo	501	0.074332	62095
4	Andaman & Nicobar Islands	2018	1	OnePlus	332	0.049258	62095

In [65]:

```
device_merge["Brand_Usage_Ratio"] = (
    device_merge["Registered Users_x"] /
    device_merge["Registered Users_y"]
)

device_merge[["State","Brand","Brand_Usage_Ratio"]]
```

Out[65]:

	State	Brand	Brand_Usage_Ratio
0	Andaman & Nicobar Islands	Xiaomi	0.026814
1	Andaman & Nicobar Islands	Samsung	0.023271
2	Andaman & Nicobar Islands	Vivo	0.015814
3	Andaman & Nicobar Islands	Oppo	0.008068
4	Andaman & Nicobar Islands	OnePlus	0.005347
5	Andaman & Nicobar Islands	Realme	0.005089
6	Andaman & Nicobar Islands	Apple	0.003688
7	Andaman & Nicobar Islands	Motorola	0.003640
8	Andaman & Nicobar Islands	Lenovo	0.003253
9	Andaman & Nicobar Islands	Huawei	0.002544
10	Andaman & Nicobar Islands	Others	0.011015
11	Andaman & Nicobar Islands	Xiaomi	0.037088
12	Andaman & Nicobar Islands	Samsung	0.030969
13	Andaman & Nicobar Islands	Vivo	0.024527
14	Andaman & Nicobar Islands	Oppo	0.011402
15	Andaman & Nicobar Islands	OnePlus	0.007505
16	Andaman & Nicobar Islands	Realme	0.007489
17	Andaman & Nicobar Islands	Motorola	0.005137
18	Andaman & Nicobar Islands	Apple	0.004960
19	Andaman & Nicobar Islands	Lenovo	0.003929
20	Andaman & Nicobar Islands	Huawei	0.003704
21	Andaman & Nicobar Islands	Others	0.014752
22	Andaman & Nicobar Islands	Xiaomi	0.047508
23	Andaman & Nicobar Islands	Samsung	0.038795
24	Andaman & Nicobar Islands	Vivo	0.033223
25	Andaman & Nicobar Islands	Oppo	0.015911
26	Andaman & Nicobar Islands	OnePlus	0.009936
27	Andaman & Nicobar Islands	Realme	0.009711
28	Andaman & Nicobar Islands	Motorola	0.006281
29	Andaman & Nicobar Islands	Apple	0.005975
30	Andaman & Nicobar Islands	Huawei	0.005057
31	Andaman & Nicobar Islands	Lenovo	0.004815
32	Andaman & Nicobar Islands	Others	0.018439

	State	Brand	Brand_Usage_Ratio
33	Andaman & Nicobar Islands	Xiaomi	0.059892
34	Andaman & Nicobar Islands	Samsung	0.047282
35	Andaman & Nicobar Islands	Vivo	0.044158
36	Andaman & Nicobar Islands	Oppo	0.020420
37	Andaman & Nicobar Islands	Realme	0.012239
38	Andaman & Nicobar Islands	OnePlus	0.011611
39	Andaman & Nicobar Islands	Motorola	0.007633
40	Andaman & Nicobar Islands	Apple	0.007327
41	Andaman & Nicobar Islands	Huawei	0.006410
42	Andaman & Nicobar Islands	Lenovo	0.006071
43	Andaman & Nicobar Islands	Others	0.022095
44	Andaman & Nicobar Islands	Xiaomi	0.073694
45	Andaman & Nicobar Islands	Samsung	0.057444
46	Andaman & Nicobar Islands	Vivo	0.055077
47	Andaman & Nicobar Islands	Oppo	0.025767
48	Andaman & Nicobar Islands	Realme	0.014446
49	Andaman & Nicobar Islands	OnePlus	0.013769
50	Andaman & Nicobar Islands	Motorola	0.009035
51	Andaman & Nicobar Islands	Apple	0.008294
52	Andaman & Nicobar Islands	Huawei	0.008294
53	Andaman & Nicobar Islands	Lenovo	0.006860
54	Andaman & Nicobar Islands	Others	0.026798
55	Andaman & Nicobar Islands	Xiaomi	0.085289
56	Andaman & Nicobar Islands	Samsung	0.067075
57	Andaman & Nicobar Islands	Vivo	0.066060
58	Andaman & Nicobar Islands	Oppo	0.030679
59	Andaman & Nicobar Islands	Realme	0.016298
60	Andaman & Nicobar Islands	OnePlus	0.015299
61	Andaman & Nicobar Islands	Huawei	0.010371
62	Andaman & Nicobar Islands	Motorola	0.010146
63	Andaman & Nicobar Islands	Apple	0.009163
64	Andaman & Nicobar Islands	Lenovo	0.007682
65	Andaman & Nicobar Islands	Others	0.031903

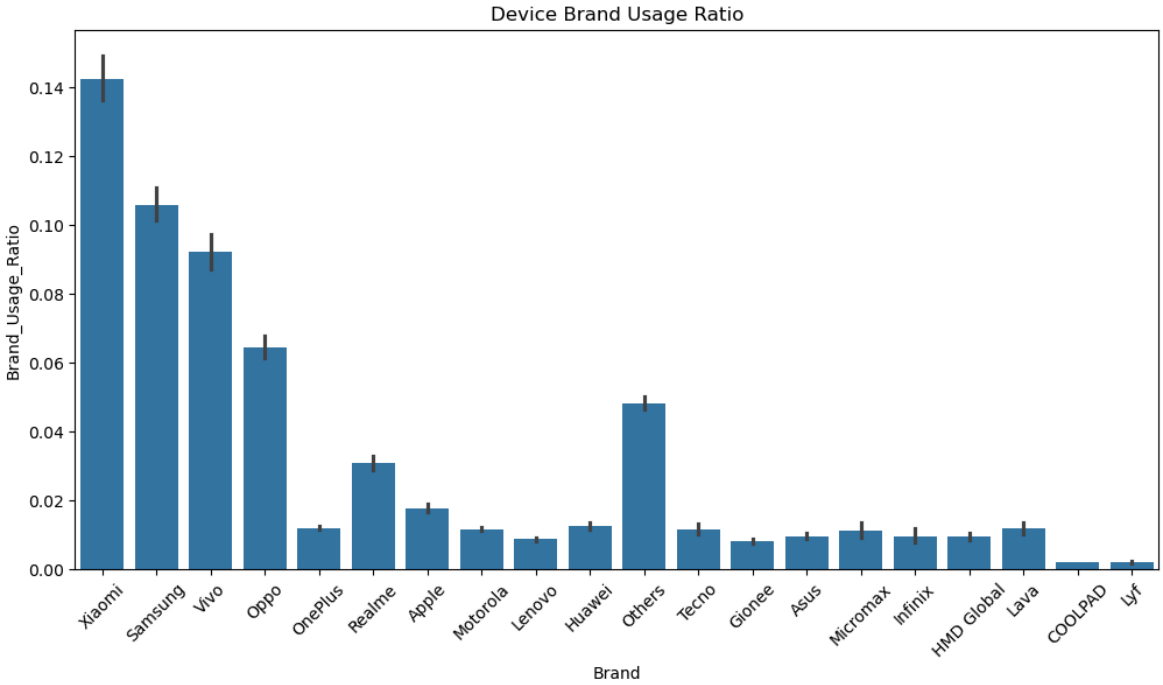
	State	Brand	Brand_Usage_Ratio
66	Andaman & Nicobar Islands	Xiaomi	0.093711
67	Andaman & Nicobar Islands	Vivo	0.075449
68	Andaman & Nicobar Islands	Samsung	0.072598
69	Andaman & Nicobar Islands	Oppo	0.033948
70	Andaman & Nicobar Islands	Realme	0.018649
71	Andaman & Nicobar Islands	OnePlus	0.016829
72	Andaman & Nicobar Islands	Huawei	0.011676
73	Andaman & Nicobar Islands	Motorola	0.010629
74	Andaman & Nicobar Islands	Apple	0.010146
75	Andaman & Nicobar Islands	Lenovo	0.007778
76	Andaman & Nicobar Islands	Others	0.033980
77	Andaman & Nicobar Islands	Xiaomi	0.100797
78	Andaman & Nicobar Islands	Vivo	0.083582
79	Andaman & Nicobar Islands	Samsung	0.078911
80	Andaman & Nicobar Islands	Oppo	0.037507
81	Andaman & Nicobar Islands	Realme	0.021403
82	Andaman & Nicobar Islands	OnePlus	0.018101
83	Andaman & Nicobar Islands	Huawei	0.012851
84	Andaman & Nicobar Islands	Motorola	0.010983
85	Andaman & Nicobar Islands	Apple	0.010919
86	Andaman & Nicobar Islands	Lenovo	0.007859
87	Andaman & Nicobar Islands	Others	0.035156
88	Andaman & Nicobar Islands	Xiaomi	0.109558
89	Andaman & Nicobar Islands	Vivo	0.094694
90	Andaman & Nicobar Islands	Samsung	0.085917
91	Andaman & Nicobar Islands	Oppo	0.041179
92	Andaman & Nicobar Islands	Realme	0.024334
93	Andaman & Nicobar Islands	OnePlus	0.019068
94	Andaman & Nicobar Islands	Huawei	0.014349
95	Andaman & Nicobar Islands	Apple	0.011837
96	Andaman & Nicobar Islands	Motorola	0.011515
97	Andaman & Nicobar Islands	Lenovo	0.008052
98	Andaman & Nicobar Islands	Others	0.036348

	State	Brand	Brand_Usage_Ratio
99	Andaman & Nicobar Islands	Xiaomi	0.125695
100	Andaman & Nicobar Islands	Vivo	0.114695
101	Andaman & Nicobar Islands	Samsung	0.097109
102	Andaman & Nicobar Islands	Oppo	0.047927
103	Andaman & Nicobar Islands	Realme	0.028360
104	Andaman & Nicobar Islands	OnePlus	0.020469
105	Andaman & Nicobar Islands	Huawei	0.016185
106	Andaman & Nicobar Islands	Apple	0.012755
107	Andaman & Nicobar Islands	Motorola	0.012545
108	Andaman & Nicobar Islands	Lenovo	0.008890
109	Andaman & Nicobar Islands	Others	0.042145
110	Andaman & Nicobar Islands	Xiaomi	0.157726
111	Andaman & Nicobar Islands	Vivo	0.153893
112	Andaman & Nicobar Islands	Samsung	0.121942
113	Andaman & Nicobar Islands	Oppo	0.060069
114	Andaman & Nicobar Islands	Realme	0.036798
115	Andaman & Nicobar Islands	OnePlus	0.023190
116	Andaman & Nicobar Islands	Huawei	0.019535
117	Andaman & Nicobar Islands	Motorola	0.014607
118	Andaman & Nicobar Islands	Apple	0.014011
119	Andaman & Nicobar Islands	Lenovo	0.009759
120	Andaman & Nicobar Islands	Others	0.054143
121	Andaman & Nicobar Islands	Xiaomi	0.199662
122	Andaman & Nicobar Islands	Vivo	0.197681
123	Andaman & Nicobar Islands	Samsung	0.160045
124	Andaman & Nicobar Islands	Oppo	0.076061
125	Andaman & Nicobar Islands	Realme	0.048249
126	Andaman & Nicobar Islands	OnePlus	0.026379
127	Andaman & Nicobar Islands	Huawei	0.023706
128	Andaman & Nicobar Islands	Motorola	0.016620
129	Andaman & Nicobar Islands	Apple	0.015670
130	Andaman & Nicobar Islands	Lenovo	0.011402
131	Andaman & Nicobar Islands	Others	0.068331

	State	Brand	Brand_Usage_Ratio
5478	West Bengal	Xiaomi	0.178721
5479	West Bengal	Samsung	0.141018
5480	West Bengal	Vivo	0.085429
5481	West Bengal	Oppo	0.067897
5482	West Bengal	Realme	0.041494
5483	West Bengal	Motorola	0.018438
5484	West Bengal	Lenovo	0.017439
5485	West Bengal	Asus	0.013202
5486	West Bengal	Huawei	0.011762
5487	West Bengal	Micromax	0.009682
5488	West Bengal	Others	0.080148
5489	West Bengal	Xiaomi	0.195232
5490	West Bengal	Samsung	0.156842
5491	West Bengal	Vivo	0.097404
5492	West Bengal	Oppo	0.076888
5493	West Bengal	Realme	0.047384
5494	West Bengal	Motorola	0.019684
5495	West Bengal	Lenovo	0.018191
5496	West Bengal	Asus	0.014205
5497	West Bengal	Huawei	0.012834
5498	West Bengal	Micromax	0.010207
5499	West Bengal	Others	0.086506
5500	West Bengal	Xiaomi	0.210927
5501	West Bengal	Samsung	0.172077
5502	West Bengal	Vivo	0.110412
5503	West Bengal	Oppo	0.085659
5504	West Bengal	Realme	0.053975
5505	West Bengal	Motorola	0.020764
5506	West Bengal	Lenovo	0.018976
5507	West Bengal	Asus	0.015051
5508	West Bengal	Huawei	0.013756
5509	West Bengal	Micromax	0.010587
5510	West Bengal	Others	0.091398

	State	Brand	Brand_Usage_Ratio
5511	West Bengal	Xiaomi	0.226382
5512	West Bengal	Samsung	0.187186
5513	West Bengal	Vivo	0.122735
5514	West Bengal	Oppo	0.094003
5515	West Bengal	Realme	0.060943
5516	West Bengal	Motorola	0.021705
5517	West Bengal	Lenovo	0.019724
5518	West Bengal	Asus	0.015782
5519	West Bengal	Huawei	0.014622
5520	West Bengal	Micromax	0.010926
5521	West Bengal	Others	0.097078
5522	West Bengal	Xiaomi	0.242326
5523	West Bengal	Samsung	0.199579
5524	West Bengal	Vivo	0.135772
5525	West Bengal	Oppo	0.101750
5526	West Bengal	Realme	0.068244
5527	West Bengal	Motorola	0.022259
5528	West Bengal	Lenovo	0.019848
5529	West Bengal	Asus	0.016896
5530	West Bengal	Huawei	0.015311
5531	West Bengal	Micromax	0.011049
5532	West Bengal	Others	0.101827
5533	West Bengal	Xiaomi	0.258886
5534	West Bengal	Samsung	0.210147
5535	West Bengal	Vivo	0.150357
5536	West Bengal	Oppo	0.109572
5537	West Bengal	Realme	0.078925
5538	West Bengal	Motorola	0.022401
5539	West Bengal	Lenovo	0.019513
5540	West Bengal	Asus	0.016551
5541	West Bengal	Huawei	0.015182
5542	West Bengal	Apple	0.011201
5543	West Bengal	Others	0.107267

In [66]:
```
plt.figure(figsize=(12,6))

sns.barplot(
    data=device_merge,
    x="Brand",
    y="Brand_Usage_Ratio"
)

plt.xticks(rotation=45)

plt.title("Device Brand Usage Ratio")

plt.show()
```



Task 4 Summary

This section combined multiple datasets to generate deeper insights.

These insights can help digital payment platforms optimize their market strategies.

Task 5: Data Visualization

Data visualization helps transform raw data into meaningful graphical insights.

In this section we create visualizations to understand:

- Transaction trends over time
- Distribution of transaction types
- Population density across districts

Task 5.1 Transaction Trend for a Selected State

We analyze how the number of transactions and transaction amount change over time for a specific state.

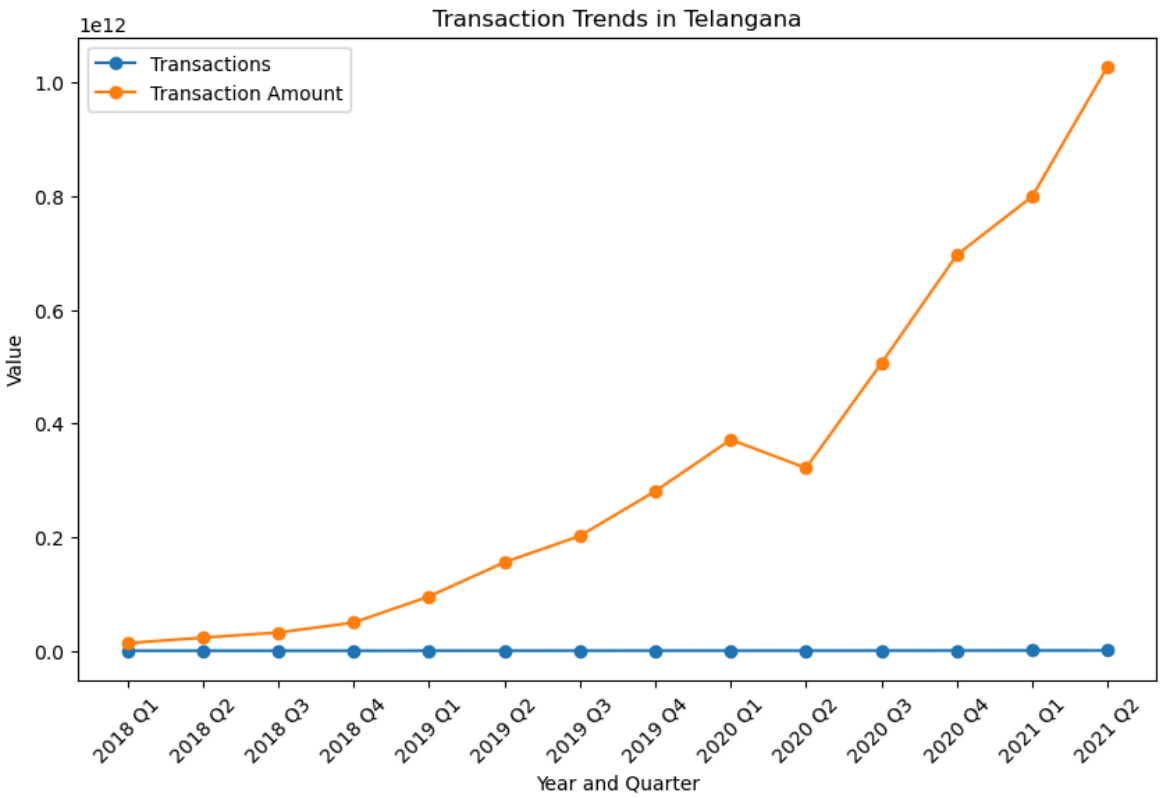
In [67]:
```
selected_state = "Telangana"

state_data = state_txn_users[state_txn_users["State"] == selected_state]

trend_data = state_data.groupby(["Year","Quarter"])[
    ["Transactions","Amount (INR)"]
].sum().reset_index()

trend_data["Time"] = trend_data["Year"].astype(str) + " Q" + trend_data["Quarter

trend_data
```

Out[67]:

	Year	Quarter	Transactions	Amount (INR)	Time
0	2018	1	8522766	1.330820e+10	2018 Q1
1	2018	2	10764670	2.301720e+10	2018 Q2
2	2018	3	19332703	3.203009e+10	2018 Q3
3	2018	4	28921154	4.950909e+10	2018 Q4
4	2019	1	65290702	9.552326e+10	2019 Q1
5	2019	2	89610379	1.556764e+11	2019 Q2
6	2019	3	121495376	2.019300e+11	2019 Q3
7	2019	4	163572258	2.807468e+11	2019 Q4
8	2020	1	199409525	3.713438e+11	2020 Q1
9	2020	2	152055143	3.215318e+11	2020 Q2
10	2020	3	236372126	5.059611e+11	2020 Q3
11	2020	4	343643787	6.963504e+11	2020 Q4
12	2021	1	412572110	7.993593e+11	2021 Q1
13	2021	2	495867544	1.027958e+12	2021 Q2

In [68]:
```
plt.figure(figsize=(10,6))

plt.plot(trend_data["Time"], trend_data["Transactions"], marker='o', label="Tran

plt.plot(trend_data["Time"], trend_data["Amount (INR)"], marker='o', label="Tran

plt.xticks(rotation=45)

plt.title("Transaction Trends in Telangana")

plt.xlabel("Year and Quarter")

plt.ylabel("Value")

plt.legend()
```

```
plt.show()
```



Interpretation

The visualization highlights the growth pattern of digital transactions in the selected state over time.

Increasing trends indicate growing digital payment adoption and higher transaction volumes.

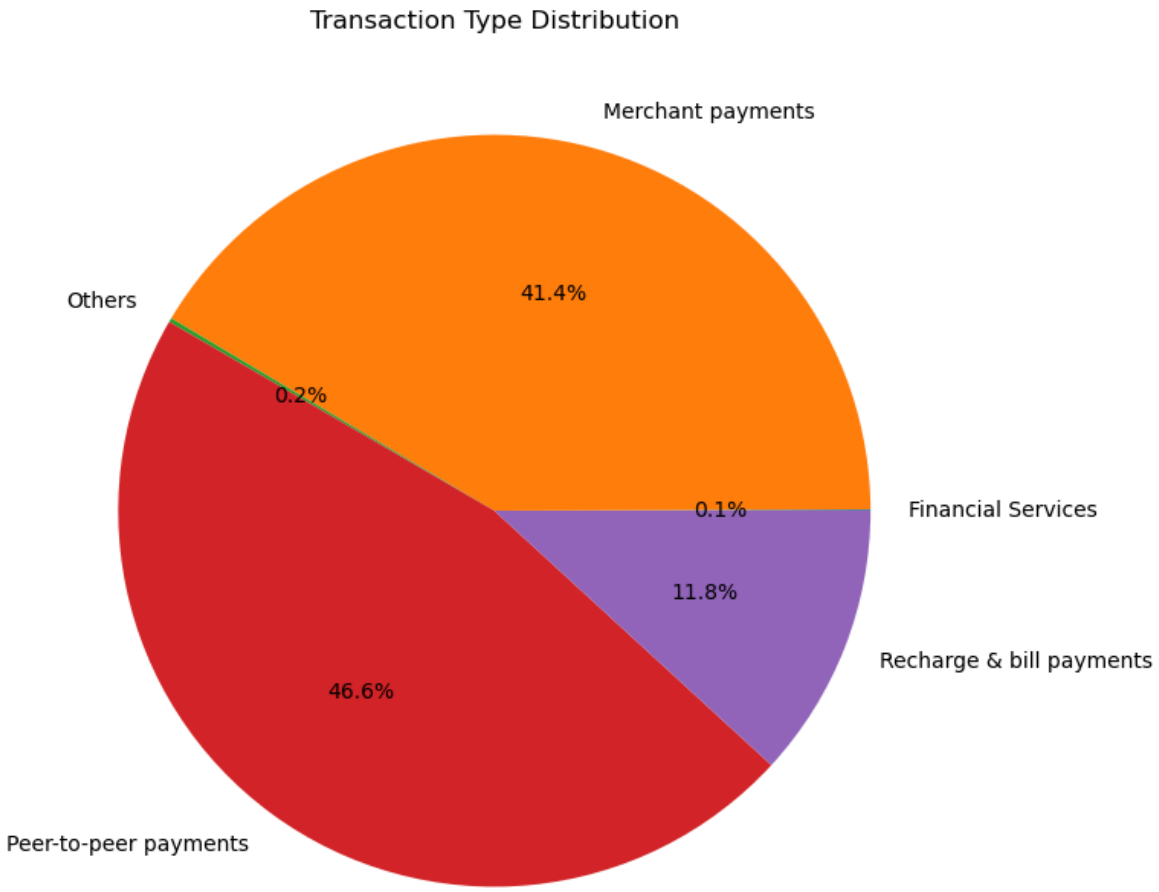
Task 5.2 Distribution of Transaction Types

A pie chart is created to show the proportion of each transaction category in the selected state and quarter.

In [69]:
```
selected_year = latest_year
selected_quarter = latest_quarter

txn_data = state_txn_split[
    (state_txn_split["State"] == selected_state) &
    (state_txn_split["Year"] == selected_year) &
    (state_txn_split["Quarter"] == selected_quarter)
]

txn_summary = txn_data.groupby("Transaction Type")["Transactions"].sum()
```

In [70]:
```
plt.figure(figsize=(8,8))

plt.pie(
    txn_summary,
```

```
    labels=txn_summary.index,
    autopct='%1.1f%%'
)

plt.title("Transaction Type Distribution")

plt.show()
```



Interpretation

The pie chart shows the share of each transaction category.

Peer-to-peer payments and bill payments typically dominate digital payment ecosystems.

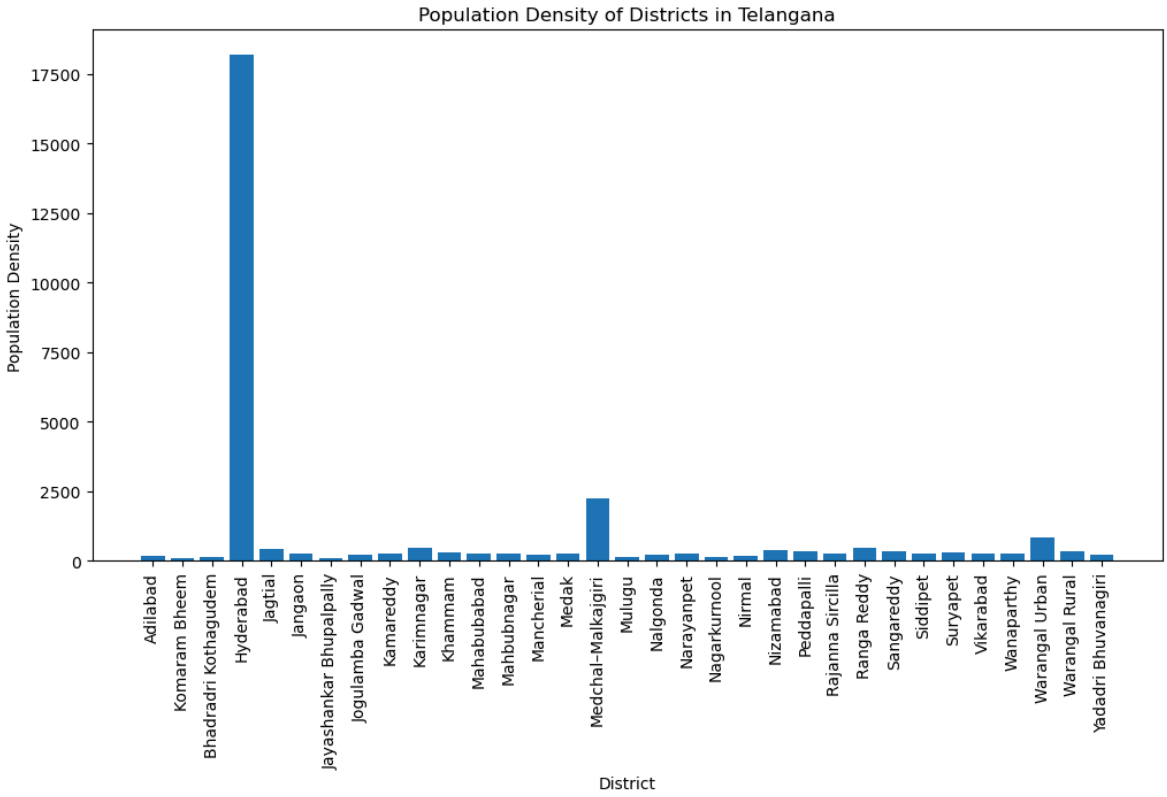
Task 5.3 Population Density Visualization

We visualize population density across districts within a selected state.

In [72]:
```
density_data = district_demographics[
    district_demographics["State"] == selected_state
]
```

In [73]:
```
plt.figure(figsize=(12,6))

plt.bar(
    density_data["District"],
```

```
    density_data["Density"]
)

plt.xticks(rotation=90)

plt.title("Population Density of Districts in Telangana")

plt.xlabel("District")

plt.ylabel("Population Density")

plt.show()
```



Interpretation

The bar chart illustrates population density differences across districts.

Higher density districts may have greater digital payment activity due to urbanization.

Task 6.1 Trends and Patterns in Transaction Data

From the analysis performed in previous sections, several trends can be observed:

1. Digital transaction volumes have steadily increased across most states over time.
2. Urbanized states tend to exhibit higher transaction volumes.
3. Peer-to-peer payments are the most dominant transaction category.
4. Smartphone penetration plays a crucial role in digital payment adoption.

These patterns highlight the growing importance of mobile-based digital payment platforms in India.

Task 6.2 Demographic and Transaction Correlation

The analysis of population density and transaction volumes indicates that:

• Districts with higher population density generally exhibit higher transaction activity.

• Urban districts tend to have greater digital payment adoption.

• Rural regions may have lower transaction volumes due to limited digital infrastructure.

This demonstrates that demographic characteristics play a significant role in digital payment adoption.

Task 6.3 Key Insights and Recommendations

Key Insights

1. Digital payment adoption is increasing across India.
2. Certain states contribute disproportionately to transaction volume.
3. Population density and urbanization influence transaction behavior.
4. Smartphone usage strongly correlates with digital payment adoption.

Recommendations

1. Payment platforms should focus on expanding digital infrastructure in rural districts.
2. Partnerships with smartphone manufacturers can increase payment app adoption.
3. Awareness campaigns can help increase digital payment usage in low adoption states.
4. Data-driven targeting can help identify high growth regions for digital payment expansion.

Final Summary

This case study analyzed PhonePe digital transaction data along with demographic information across states and districts in India.

Key activities performed during the analysis include:

• Data loading and exploration
• Data quality validation
• Transaction trend analysis
• Device usage analysis
• Demographic correlation analysis
• Data visualization

The analysis revealed important patterns in digital payment adoption and highlighted how demographic and technological factors influence transaction behavior.

Overall, the study demonstrates the rapid growth of digital payments in India and identifies opportunities to further expand digital payment adoption.

Final Project Conclusion

This project analyzed PhonePe digital transaction data across India using Python. The analysis explored transaction growth trends, digital payment adoption across states, device usage patterns, and demographic influences on transactions. By combining transaction, user, and population datasets, I identified key insights such as higher digital adoption in urban regions, strong growth in peer-to-peer transactions, and the influence of smartphone penetration on digital payments.

In [ ]: